QUALIFICATION STANDARD AND PLAN (QSP)

NAVAL ENGINEERING COMMON

MITE ID
Ashore Phase: 119913
Afloat Phase: 119912

Regular Force: Naval Combat Engineering (00344)
Marine Systems Engineering (00345)
Training Authority (TA): Commander Naval Training Group (CNTG)
Curriculum Control Authority (CCA): HMCS Venture

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RECORD OF AMENDMENTS

1. This QSP is developed under the authority of the Royal Canadian Navy (RCN). RCN, through its Training Authority (TA), Director Maritime Training and Education (DMTE) has delegated “authority” of this QSP to the Curriculum Control Authority (CCA), Canadian Forces Naval Engineering School Halifax (CFNES). The approval/revision date should also appear in the footer of every page as an additional means of QSP verification. Amendments detailed in Annexes D and/or L.

Revision	Change Description	Date	Authority
1.0	QSPRB	17 Jun 13	<<<O.S.B.>>> Cdr D.J. Benoit CFNES Cmdt
2.0	New RCN Format	27 Jul 23	CO HMCS Venture

NOTE:

Personnel programming courses must ensure that their hard copy of this document is current and identical to the electronic copy on the QSP SharePoint Site. See above for current changes.

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FOREWORD

1. This QSP is issued on the authority of the Commander of the Royal Canadian Navy and managed by the Commander Naval Personnel and Training Group (CNPTG) on behalf of the Assistant Chief of Naval Staff Personnel and Training (A/CNS P&T).

2. This QSP is based upon the following references:

   a. A-P9 – 050 Series, Canadian Forces Manual of Individual Training and Education (CFITES);

   b. A-PD-055-002/PP-003 Canadian Forces Officer General Specification;

   c. A-PD-055-002/PP-002 Non-Commissioned Member General Specification;

   d. A-PD-055-002/PP-001 NCS Eng Job Based Occupational Specification;

   e. A-PD-055-002/PP-001 MS ENG MOSID 00345 Job Based Occupation Specification for Marine Systems Engineering Officer and

   f. NAVORD 4500-0 - Structured Delegation, and NPTORD 4500-22 – Qualification Standard and Plan Production and Management.

3. This QSP publication is effective on receipt and supersedes Naval Engineering Common QSP revision 1.0, dated 13 June 2013. This QSP is designed to assist in developing, conducting, evaluating, and validating the Individual Training and Education (IT&E) required for training.

4. The QS and TP at Chapters 1, 2, 3, and 4 were developed by the Qualification Standard Writing Board (QSWB) convened 13 February 2012 at CFNES Halifax working in collaboration with subject matter experts drawn from participating organizations. All aspects of this document were prepared in accordance with the concept of training outlined in A-P9-050 Series, Canadian Forces Manual of Individual Training and Education. Full details of the WBs are contained within Annex D and L.

5. The criteria for major and minor amendments, and the approval process, are identified in NPTORD 4500-22.

6. In circumstances where the QS and/or TP cannot be adhered to, submission of a Training Waiver in accordance with NPTORD 4500-22 is required.

QUALIFICATION STANDARD

CHAPTER 1 – QUALIFICATION STANDARD

AIM

1. The aim of this QSP is to establish the common training requirements between NCS and MS Eng Officers in order to continue their training to achieve basic occupation qualification.

GRANTING OF NQUAL

2. A trainee will achieve the qualification Naval Engineering Common Ashore (MITE 119913) and Naval Engineering Common Afloat (MITE 119912) when the standards listed in the Performance Objectives (POs) have been met.

CO-QUALIFICATION

3. The NEC OJPR shall be completed concurrently with the NCS Eng or the MS Eng Basic Qualification Officer Afloat OJPR. The expected time to complete the OJT is 9 months. If a candidate is required to attend career coursing during the afloat period, the duration of the afloat period is to be extended by the period of time spent on this course. If the trainee has not completed Officer of the Day (OOD) training, an additional 3 months will be allotted to complete the OOD OJPR.

4. The candidate's Commanding Officer (CO) may grant an extension IAW NPTORD 4500-55.

PREREQUISITES

5. To be select ed for this course the loading authority must ensure that trainees meet the following prerequisites:

   a. Be a Marine Systems Engineer or Combat Systems Engineer;

   b. Have completed or have been waived the applicable Academic Bridging course if they are a Commissioned From the Ranks Program (CFRP) entry officers; and

   c. Have completed or have been waived the Naval Engineering Indoctrination (NEI) course (MITE 109286).

CHAPTER 2 – PERFORMANCE OBJECTIVES

PO 001- DESCRIBE OPERATIONAL READINESS

1. Performance: Describe Operational Readiness

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) References; and

      (3) Engineering and Operational Staff Support.

   b. Denied: N/A

   c. Environment:

      (1) Ashore; and

      (2) Afloat.

3. Standard: The NTO shall describe Operational Readiness:

   a. IAW refs, by :

      (1) Describing Readiness Policies and Programs; and

      (2) Describing Ship's States of Readiness.

4. Reference:

   a. A1, A9, A10, A11, A14, A15, A25, A26, A29, A32, A33, A37, A40, A42, A49, and A55.

5. Task Numbers: Nil.

6. Remarks: Nil.

PO 002- DESCRIBE RISK MANAGEMENT

1. Performance: Describe Risk Management

2. Conditions:

    a. Given:

        (1) Operational Ship;

        (2) References; and

        (3) Engineering and Operational Staff Support.

    b. Denied: N/A

    c. Environment:

        (1) Ashore; and

        (2) Afloat.

3. Standard: The NTO shall describe Risk Management:

    a. IAW refs, by :

        (1) Identifying elements of Risk Management; and

        (2) Describing the Risk Management Framework.

4. Reference:

    a. A1, A6, A12, A13, A14, A15, A18, A20, A33, A37, A52, A58, A62, A64, A65, A66, A67, A68, A69, A73, A74, A75, and A76.

5. Task Numbers: Nil.

6. Remarks: Nil.

PO 003- DESCRIBE NAVAL TESTS AND TRIALS

1. Performance: Describe Naval Tests and Trials
2. Conditions:
    a. Given:
        (1) Operational Ship;
        (2) References;
        (3) Engineering and Operational Staff Support;
        (4) Trial Staff; and
        (5) Ranges, Test Facilities and test equipment.
    b. Denied: N/A
    c. Environment:
        (1) Ashore; and
        (2) Afloat.
3. Standard: The NTO shall describe Naval Tests and Trials IAW refs.
4. Reference:
    a. A1, A14, A15, A28, A37, A38, A42, A49,
5. Task Numbers: Nil.
6. Remarks: Nil.

PO 004- SUPPORT ENGINEERING ANALYSIS

1. Performance: Support Engineering Analysis

2. Conditions:

    a. Given:

        (1) Operational Ship;

        (2) References; and

        (3) Engineering and Operational Staff Support.

    b. Denied: N/A

    c. Environment:

        (1) Ashore; and

        (2) Afloat.

3. Standard: The NTO shall Support Engineering Analysis:

    a. IAW refs, by :

        (1) Supporting Failure Investigations; and

        (2) Describing Engineering Analysis Tools.

4. Reference:

    a. A1, A3, A6, A12, A14, A15, A16, A17, A18, A22, A24, A28, A30, A33, A37, A38, A42, A48, A52, and A53.

5. Task Numbers: Nil.

6. Remarks: Nil.

PO 005- DESCRIBE TRAINING AND PERSONNEL MANAGEMENT

1. Performance: Describe Training and Personnel Management

2. Conditions:

    a. Given:

        (1) Operational Ship;

        (2) References; and

        (3) Engineering and Operational Staff Support.

    b. Denied: N/A

    c. Environment:

        (1) Ashore; and

        (2) Afloat.

3. Standard: The NTO shall describe Training and Personnel Management:

    a. IAW refs, by :

        (1) Describing Training of Naval Personnel; and

        (2) Describing Personnel Management.

4. Reference:

    a. A1, A2, A6, A7, A22, A25, A26, A32, A33, A34, A37, A38, A53, A55, A58, A62, and A78.

5. Task Numbers: Nil.

6. Remarks: Nil.

PO 006- DESCRIBE NAVAL MATERIAL MANAGEMENT

1. Performance: Describe Naval Material Management

2. Conditions:

   a. Given:

      (1) Operational Ship;

      (2) References; and

      (3) Engineering and Operational Staff Support.

   b. Denied: N/A

   c. Environment:

      (1) Ashore; and

      (2) Afloat.

3. Standard: The NTO shall describe Naval Material Management:

   a. IAW refs, by :

      (1) Describing Naval Maintenance Policy;

      (2) Describing Naval Maintenance Organizations;

      (3) Describing Naval Maintenance Procedures;

      (4) Describing Naval Maintenance Documentation;

      (5) Describing Life Cycle Materiel Management (LCMM);

      (6) Describing Naval Configuration Management;

      (7) Describing Engineering Quality Management; and

      (8) Describing Naval Material Assurance

4. Reference:

   a. A2, A3, A14, A15, A16, A17, A21, A23, A25, A28, A29, A30, A31, A32, A38, and A39.

5. Task Numbers: Nil.

6. Remarks: Nil.

PO 007- DESCRIBE BUSINESS MANAGEMENT

1. Performance: Describe Business Management
2. Conditions:
    a. Given:
        (1) Operational Ship;
        (2) References; and
        (3) Engineering and Operational Staff Support.
    b. Denied: N/A
    c. Environment:
        (1) Ashore; and
        (2) Afloat.
3. Standard: The NTO shall describe Business Management:
    a. IAW refs, by :
        (1) Describing Contract Management; and
        (2) Describing Capability Planning.
4. Reference:
    a. A2 and A6.
5. Task Numbers: Nil.
6. Remarks: Nil.

PO 008- MANAGE PROJECTS

1. Performance: Manage Projects
2. Conditions:
    a. Given:
        (1) Operational Ship;
        (2) References; and
        (3) Engineering and Operational Staff Support.
    b. Denied: N/A
    c. Environment:
        (1) Ashore; and
        (2) Afloat.
3. Standard: The NTO shall Manage Projects:
    a. IAW refs, by :
        (1) Describing Project Management Principles;
        (2) Describing the Defence Services Program; and
        (3) Conducting Engineering Design and Integration.
4. Reference:
    a. A2, A7, A14, A15, A38, and A42.
5. Task Numbers: Nil.
6. Remarks: Nil.

TRAINING PLAN

CHAPTER 3 – TRAINING MANAGEMENT DETAILS

TRAINING STRATEGY

1. The following training strategies were selected for this course:

   a. The ashore phase of training which will be delivered through a series of lectures and tours; and

   b. The afloat phase of training which requires self-study and mentorship from the CSE/MSE departments of the candidate's ship.

TRAINING DURATION

2. Based on existing resources and approved instructional design requirements, the approved Course Design authorizes 27.5 training days.

REQUIRED COURSE STAFFING

3. The following key personnel are required to conduct the training:

REQUIRED COURSE STAFFING				
TITLE	**RANK**	**QUALIFICATION**	**QTY**	**REMARKS**
Instructor	Lt(N) / LCdr/ civilian	1) HOD qualified or equivalent	3	

TRAINING LIMITATION

4. A training limitation may be due to constraints of resources, personnel or equipment. A training limitation is the difference between what a qualified individual must be able to do and what training can prepare them to do. The following limitations are identified below:

   a. Nil

COURSE CAPACITY

5. The capacity per session is based on the guidelines in CFITES Vol 5 – Instructor to Learner Ratios:

   a. Maximum Load – 50; and

b. Minimum – 16.

DESIGNATED SCHEDULING AND COURSE LOADING

6. Training will be scheduled utilizing the HRMS Training (MITE) module to meet the training requirements (IAW NPTORD 4540-2). The length of the course in MITE, is to be based on the course duration (see table #2) plus planned annual leave, training breaks, training system holiday, and weekends.

7. The authority to load members onto this course will be IAW NPTORD 4540-43. NPTG HQ is to be included as an info addressee on all loading messages. Course loading will be accomplished by:

a. NFS Campus Support

QUALIFICATION AND ASSESSMENT

8. This is a pass/fail course. Students must successfully pass the PC and critical EC to achieve the qualification.

9. In order to achieve a passing mark for the NEC ashore phase, trainees must achieve a minimum mark of 60%. The NEC ashore phase is assessed by a critical EC and an EC. Trainees must achieve a minimum mark of 60% in the critical EC and EC.

10. The final mark for NEC ashore phase is calculated using the following formula:

a. Total Course Mark = 0.7x(Critical EC Mark) + 0.3x(EC Mark)

11. The marking scheme for the NEC ashore phase, including Annex K – Assessment Plan, shall be provided to trainees prior to the start of the NEC course.

12. During the ashore phase, any candidate who fails the critical EC shall be given a supplemental. IAW CFNES SO 425 (Training Review Boards), a TRB shall be convened whenever a student fails a supplemental critical EC. A TRB may also be convened for other reasons if deemed necessary IAW MARCORD 9-60.

13. As specified in Chapter 1, upon completion of the NEC afloat phase and the BOQ afloat phase, an oral examination board (PC 001) will be convened for each NTO Trainee IAW MARCORD 9-20. Board examinable subjects are based on the relevant OJPR, NEC QSP EOs, and the applicable NCS Eng or MS Eng BOQ QSP EOs. The candidate shall achieve a minimum overall average mark of 60%, including a minimum of 60% for each of questions 1 and 2 (critical questions). Details of the oral examination

14. Board and the marking guide can be found in Annex L. Upon successful completion of the oral examination board, the NTO Trainee will be awarded the NCS ENG occupation qualification (AIRX) or the MS ENG occupation qualification (AIRY), as applicable. Any candidate who fails PC 001 shall be given a supplemental PC.If the trainee fails the

supplemental PC, a TRB shall be convened IAW NPTORD 4500-54 – Training Review Board Process..

15. Students will be monitored and assessed both on performance and conduct throughout the course. Counselling, progress monitoring, and maintenance of student files will be in accordance with NPTORD 4500-51. Although not mandated specifically in the POs, the general demands of service members and the leadership demands of junior and senior leaders form an integral part of individual training. As such, students are presumed to possess the leader and/or follower attributes commensurate with their rank.

UNSATISFACTORY PROGRESS

16. Any student who fails to successfully complete all POs in the allocated number of attempts will be referred to the Commandant (Cmdt) or Commanding Officer (CO) of the training establishment (TE) to determine the requirement for a TRB IAW NPTORD 4500-54.

REMOVAL FROM TRAINING

17. A TE CO/Cmdt may direct that a student be removed from training for reasons of deficiency in performance, conduct (discipline, attitude, teamwork, safety, leadership), missed training, medical, and other non-performance/conduct related issues. In these cases, Cease Training Procedures are to be strictly adhered to IAW NPTORD 4500-29 or 4500-54.

RECERTIFICATION REQUIREMENTS

18. There are no additional requirements to maintain currency in this qualification.

LANGUAGE OF INSTRUCTION

19. The language of instruction shall be in accordance with the Delivery of Training and Education in Both Official Languages (DAOD 5039-6).

PRIOR LEARNING ASSESSMENT AND RECOGNITION (PLAR)

20. PLARs shall be conducted in accordance with DAOD 5031-1 Canadian Forces Military Equivalencies Program, through the RCN PLAR submissions coordinator.

POST COURSE REVIEWS (PCR)

21. PCRs are to be conducted on all steady state courses IAW NPTORD 4500-23.

COURSE REPORTING

22. All members who successfully, unsuccessfully or partially complete this course shall be awarded a Canadian Forces Course Report (CF 377) which is included within

the applicable NTO applications course report. These are to be completed IAW DAOD 5031-9 - Course Reporting and Certificates, NTI - Course Reporting and Certificates, CMP Instruction 01/06 - Course Reporting and Certificates, CFITES Vol. 13 - Administration of Training Establishments and Programmes, and CANFORGEN 122/06 - Course Reporting and Certificates. The Course Report shall be descriptive, identifying the student's strengths and weaknesses with specific comments on any PO for which the student demonstrated outstanding performance or had difficulty achieving the standard, any conduct relevant to course performance, and any conditions or situations that affected course performance.

CHAPTER 4 - LESSON SPECIFICATIONS

PO 001 – DESCRIBE OPERATIONAL READINESS

EO 001.01 Describe Readiness Policies and Programs

1. Performance: Describe Readiness Policies and Programs.

2. Conditions:

   a. Given:

      (1) Nil;

   b. Denied:

      (1) N/A; and

   c. Environment:

      (1) Classroom.

3. Standard: The NTO shall describe Readiness Policies and Programs, including the following:

   a. Readiness and Sustainment Program (RSP);

   b. Fleet Support Resources. Capability Management;

   c. Force Generation and Employment;

   d. Shipboard Operational Readiness;

   e. Combat Readiness Requirements (CRR);

   f. Ship's Signature Management; and

   g. Explain Technical Readiness IAW the OJPR (Annex E to this document).

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Readiness and Sustainment Program (RSP)**	**Method:** Lecture **Media:** PowerPoint	**30**	**A1; A9; A10; A11; A14; A15; A25; A26; A29; A32; A33; A37; A40; A42; A49; A55**

Teaching Points		Instructional Strategies	Time (min)	References
		Environment: Classroom		
1.1	References			
1.2	Tiered Readiness Programme (TRP) 1) References; 2) Readiness Inspection (HRI, RRI); 3) Inspections; 4) Refresher Training; 5) Team Training; i. Damage Control Team Training; ii. HAZMAT Team Training; iii. Combat Team Training; iv. Weapon Certifications; v. Operational Team Training; 6) Technical Trails; 7) Directed Repair periods; and 8) Work-ups.			
2	**Fleet Support Resources. Capability Management**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**30**	
2.1	Process/Objective and link to Capability/Business Planning			
2.2	Role of DMFM (Director Maritime Fleet Management)			
2.3	Design/Development			
2.4	Acquisition			
2.5	Maintenance			
2.6	R&O			
2.7	NICP			
2.8	Change			
2.9	Disposal			
3	**Force Generation and employment**	**Method:** Lecture	**30**	

Teaching Points		Instructional Strategies	Time (min)	References
		Media: PowerPoint **Environment:** Classroom		
3.1	Doctrine and Concepts			
3.2	NDHQ to Unit level			
3.3	Fleet Operational Readiness Planning			
4	**Shipboard Operational Readiness**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**30**	
4.1	States of preparedness			
4.2	Degrees of Readiness			
5	**Combat Readiness Requirements (CRR)**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**35**	
5.1	Individual and Team Training Requirements			
5.2	Role of Sea Training			
6	**Ship's Signature Management**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**35**	
6.1	IR Signature management			
6.2	Acoustic signature management			
6.3	Magnetic signature management			
6.4	Radar cross section			
7	**Explain the policy and procedures of each of the following to the NTO Department Head (without references)**	**Method:** OJT **Media:** N/A **Environment:** Ship		
7.1	OPDEF			
7.2	MBS and Naval Ship Code			
7.3	Critical Manning Requirements			

Teaching Points		Instructional Strategies	Time (min)	References
7.4	CRRs			
7.5	Critical Supply Requirements			
7.6	Pre-sail reporting			
7.7	Watch and Station Bill			
7.8	Brief to CO			

5. Total EO Time: Total: 4 periods as follows:

   a. 4 x 50 min period – Classroom.

6. Training and Learning Aids: Nil.

7. Remarks: Guest lecturer from N37 recommended to achieve standard of para 3a, and b.

PO 001 – DESCRIBE OPERATIONAL READINESS

EO 001.02 Describe Ship's State of Readiness

1. Performance: Describe Ship's State of Readiness

2. Conditions:

   a. Given:

      (1) Nil;

   b. Denied:

      (1) N/A; and

   c. Environment:

      (1) Classroom.

3. Standard: The NTO shall describe Readiness Policies and Programs, including the following:

   a. Describing Operational Deficiencies (OPDEFS);

   b. Describing Notice for Power (NFP);

   c. Describing Personnel Readiness;

   d. Describing Scheduling;

   e. Describing Logistical Readiness;

   f. Describing Checks and Reports;

   g. Participating in Technical Readiness activities IAW the OJPR (Annex E to this document); and

   h. Preparing and staffing Technical Readiness documents IAW the OJPR (Annex E to this document).

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Operational Deficiencies (OPDEFs)**	**Method:** Lecture	**50**	**A1; A9; A10; A11; A14; A15; A25; A26; A29**

Teaching Points		Instructional Strategies	Time (min)	References
		Media: PowerPoint **Environment:** Classroom		
1.1	Purpose; 1. Link to technical assistance; and 2. Link to logistical support (e.g. Spares).			
1.2	Message Types; 1. Initial; 2. SITREP; 3. Rectified; and 4. Summary.			
1.3	Message format and completion; 1. Addresses.			
1.3	Classification; 1. Degraded/Inoperative; and 2. Categories;			
1.4	Classification; 1. Degraded/Inoperative; and 2. Categories;			
1.5	Prioritization			
1.6	Requirement			
2	**Describe Notice for Power (NFP)**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
2.1	Categories; 1. Immediate; 2. Short; and 3. Extended.			
2.2	Operational requirements			
2.3	Maintenance limitations			

Teaching Points		Instructional Strategies	Time (min)	References
3	**Describe Personnel Readiness**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
3.1	Watch and Station Bill			
3.2	Departmental Manning; 1. Minimum manning; 2. Personnel shortages; i. Personnel shortage message; ii. Critical shortage message; and iii. Training opportunity message.			
4	**Describe Scheduling**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
4.1	Formation Level 1) Operational Schedule (OPSKED); 2) Priorities; and 3) Incremental Tasking.			
4.2	Ship's Programme			
5	**Describe Logistical Readiness**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
5.1	High Priority Request (HPR)			
5.2	Emergency Requirement (EMREQ)			
5.3	Material Requirement (MATREQ)			
5.4	Transfer Requirement (TRANREQ)			

Teaching Points		Instructional Strategies	Time (min)	References
5.5	Material Accounts; 1) ESL; 2) SSS; and 3) FEL.			
6	**Describe Checks and Reports**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
6.1	Pre-sails			
6.2	Daily			
7	**Participate in the following Technical Readiness activities**	**Method:** OJT **Media:** N/A **Environment:** Ship		
7.1	Drafting of OPDEFs			
7.2	Reporting to Command			
7.3	Pre-sail checks			
7.4	Equipment faults/recommendations			
7.5	Manning issues (e.g. critical manning)			
7.6	Daily brief (e.g. rounds report)			
7.7	Development of Ship's Program/routine			
7.8	Execution of CRR (Combat readiness Requirements)			
7.9	Identification of critical supply requirements			
7.10	Creation of departmental watch and station bill			
7.11	Daily departmental rounds			
8	**Prepare and staff Technical Readiness**	**Method:** OJT **Media:** N/A **Environment:** Ship		
8.1	OPDEF Messages			
8.2	Critical Supply Requirements (HPRs, EMREQs) (with LOGO assistance)			
8.3	Personnel Shortage Manning Messages			

Teaching Points		Instructional Strategies	Time (min)	References
8.4	Routine Engineering Reports , to include: 1) OPDEF Summary Message; 2) Risk Management Records in DRMIS;			

5. Total EO Time: Total: 6 x 50 minute periods as follows:

   a. 6 x 50 min period – Classroom.

6. Training and Learning Aids: Nil.

7. Remarks: Guest lecturer from N3 recommended to achieve standard of para 3a, d, and e. Students to attend N3 morning brief in the D201 star-chamber.

PO 002 – DESCRIBE RISK MANAGEMENT

EO 002.01 Identify elements of Risk Management

1. Performance: Identify elements of Risk Management.

2. Conditions:

   a. Given:

      (1) Nil;

   b. Denied:

      (1) N/A; and

   c. Environment:

      (1) Classroom.

3. Standard: The NTO shall Identify the elements of Risk Management by:

   a. Summarizing policies and regulations;

   b. Discussing background and purpose;

   c. Identifying stakeholder organizations, roles and responsibilities;

   d. Defining issues, risk, and opportunity; and

   e. Introducing Risk Management terminology.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Summarize policies and regulations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	**A1; A6; A12; A13; A14; A15; A18; A20; A33; A37; A52; A58; A62; A64; A65; A66; A67; A68; A69; A73; A74; A75; A76**
1.1	DND/CF Integrated Risk Management Policy			
2	**Discuss background and purpose, to include the**	**Method:** Lecture **Media:** PowerPoint	**50**	

Teaching Points		Instructional Strategies	Time (min)	References
		Environment: Classroom		
2.1	History			
2.2	Risk Culture			
2.3	Risk Management in the GoC			
2.4	Risk Management in DND/CF			
2.5	Risk Management in the Navy			
2.6	Operational Risk Management			
2.7	Purpose of Risk Management			
3	**Identify stakeholders Organizations, Roles and**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
4	**Define Issue, Risk and Opportunity**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
4.1	Characteristics of Issues			
4.2	Characteristics of Risks			
4.3	Characteristics of Opportunities			
5	**Introduce Risk Management Terminology**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	

5. Total EO Time: Total: 5 x 50 minute periods as follows:

   a. 5 x 50 min period – Classroom.

6. Training and Learning Aids: Nil.

7. Remarks: N/A

PO 002 – DESCRIBE RISK MANAGEMENT

EO 002.02 Describe the Risk Management Framework

1. Performance: Describe the Risk Management Framework.

2. Conditions:

   a. Given:

      (1) Nil;

   b. Denied:

      (1) N/A; and

   c. Environment:

      (1) Classroom.

3. Standard: The NTO shall describe the Risk Management Framework, by:

   a. Describing Operational/Technical Risk Management onboard ships;

   b. Describing Integrated Risk Management;

   c. Describing Risk Management in Projects;

   d. Describing Risk Management Process; and

   e. Identifying Risk Management Enablers.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Operational/Technical Risk Management onboard ships**	**Method:** Case Study **Media:** PowerPoint / Handouts **Environment:** Classroom	**50**	**A1; A6; A12; A13; A14; A15; A18; A20; A33; A37; A52; A58; A62; A64; A65; A66; A67; A68; A69; A73; A74; A75; A76**
1.1	Identifying Risk and Risk Mitigation Measures			
2	**Describe Integrated Risk Management**	**Method:** Case Study	**50**	

Teaching Points		Instructional Strategies	Time (min)	References
		Media: PowerPoint / Handouts **Environment:** Classroom		
2.1	Risk Management Programs Onboard HMC Ships			
3	**Describe Risk Management in Projects**	**Method:** Case Study **Media:** PowerPoint / Handouts **Environment:** Classroom	**50**	
3.1	Role and Requirements of Risk Management Organizations within DND			
4	**Describe Risk Management Process**	**Method:** Case Study **Media:** PowerPoint / Handouts **Environment:** Classroom	**200**	
4.1	Risk Planning: 1. Identifying risk factors (objectives at risk); 2. Identifying risk categories; 3. Identifying probabilities and impacts (probability and impact matrix, or Risk Exposure levels); 4. Defining overall methodology; and 5. Identifying stakeholders.			
4.2	Risk Identification and methods: 1. Defining Context, Conditions, Consequences; and 2. Defining Methods.			
4.3	Risk Analysis and methods: 1. Determining attributes; 2. Defining Classification;			

Teaching Points		Instructional Strategies	Time (min)	References
	3. Perform Quantitative Risk Analysis; 4. Perform Qualitative Risk Analysis; 5. Prioritizing; and 6. Defining Accountability, Responsibility, Authority.			
4.4	Risk Response Planning and Implementation: 1. Defining Acceptance; 2. Defining Transfer; 3. Defining Mitigate (ALARP); 4. Defining Avoid;			
5	**Identify Risk Management Enablers to include the**	**Method:** Case Study **Media:** PowerPoint / Handouts **Environment:** Classroom	**50**	
5.1	Prioritization			
5.2	Risk Management Plan			
5.3	Data Collection			
5.4	Integrated Logistics Support data			
5.5	Simulation Software			
5.6	Risk Register (Risk Information Sheet)			
5.7	Risk Management Software			
5.8	Lessons Learned			
5.9	Operational and Technical Knowledge			
5.10	Communication			
5.11	Tools and decision aids			

5. Total EO Time: Total: 5 x 50 minute periods as follows:

   a. 5 x 50 min period – Classroom.

6. Training and Learning Aids: Nil.

7. Remarks: N/A

PO 003 – DESCRIBE NAVAL TESTS AND TRIALS

EO 003.01 Describe Naval Tests and Trials Policies and Programs

1. Performance: Describe Naval Tests and Trials Policies and Programs

2. Conditions:

   a. Given:

      (1) Nil;

   b. Denied:

      (1) N/A; and

   c. Environment:

      (1) Classroom.

3. Standard: The NTO shall describe Naval Tests and Trials Policies and Programs by:

   a. Describing Policies and Procedures;

   b. Describing the Trials Organization;

   c. Describing Tests and Trials;

   d. Describing the purpose, frequency, impact and considerations for major Shipboard trials;

   e. Assisting in shipboard test and trial activities IAW the OJPR (Annex E to this document);

   f. Reviewing, reporting and advising Command on results IAW the OJPR (Annex E to this document); and

   g. Explaining the policies and procedures of Naval Trials IAW the OJPR (Annex E to this document).

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Policies and procedures**	**Method:** Lecture	**15**	**A1; A14; A15; A28; A37; A38; A42; A49**

Teaching Points		Instructional Strategies	Time (min)	References
		Media: PowerPoint **Environment:** Classroom		
1.1	Trials Documentation; 1. Master Index of Trials Agenda; 2. Trials Agendas; 3. Certificates; and 4. Trial Reports.			
1.2	Trial stages; 1. Physical (alongside); 2. Functional (alongside); and 3. Operational (at sea);			
2	**Describe Trials Organization**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**10**	
2.1	Tasking Authority;			
2.2	Conducting Authority;			
2.3	FMF trials; and			
2.4	FTA.			
3	**Describe Tests and Trials**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
3.1	Purpose			
3.2	Occasions; 1. Tiered Readiness Program (TRP); 2. Periodic; 3. Post Corrective Maintenance; and 4. New Equipment.			
3.3	Types: 1. Maritime Evaluations.			
4	**Describe the purpose, frequency, impact and**	**Method:** Lecture	**100**	

Teaching Points		Instructional Strategies	Time (min)	References
	considerations for major Shipboard trials	**Media:** PowerPoint **Environment:** Classroom		
4.1	MS ENG Trials; 1. Basin Trials; 2. Main Machinery Performance Trial (MMPT); 3. Load Bank Trial; and 4. Lifting appliances certification.			
4.2	NCS ENG trials; 1. Weapons/sensor/INS alignment; 2. Acoustic; i. Static and Dynamic ranging; and ii. Sonar Performance Figures (SPFs) 3. Magnetic ranging; 4. Deperming; 5. Degaussing; and 6. Magazine Certification.			
5	**Assist in shipboard test and trial activities, to include a major at sea trial that requires external agencies**	**Method:** OJT **Media:** N/A **Environment:** Ship		
6	**Review, report and advise Command on results**	**Method:** OJT **Media:** N/A **Environment:** Ship		
7	**Explain the policies and procedures of Naval Trials**	**Method:** OJT **Media:** N/A **Environment:** Ship		
7.1	Trials Agenda			
7.2	Organization			
7.3	Ranges/Resources			

5. Total EO Time: Total: 3 periods as follows:

   a. 3 x 50 min period – Classroom.

6. Training and Learning Aids: Nil.

7. Remarks: N/A

PO 004 – SUPPORT ENGINEERING ANALYSIS

EO 004.01 Support Failure Investigations

1. Performance: Support Failure Investigations.

2. Conditions:

   a. Given:

      (1) Nil;

   b. Denied:

      (1) N/A; and

   c. Environment:

      (1) Classroom.

3. Standard: The NTO shall support Failure Investigations by:

   a. Describing Boards of Inquiry (BOI);

   b. Describing Summary Investigations (SI);

   c. Describing Informal Investigations; and

   d. Conducting Technical Investigations (TI).

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Board of Inquiry (BOI)**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**10**	**A1; A3; A6; A12; A14; A15; A16; A17; A18; A22; A24; A28; A30; A33; A37; A38; A42; A48; A52; A53**
1.1	General overview			
2	**Describe Summary Investigations (SI)**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**10**	
2.1	General overview			

Teaching Points		Instructional Strategies	Time (min)	References
3	**Describe informal investigations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**10**	
3.1	Equipment failure			
3.2	SOP Adherence			
3.3	Technical Service Call (TSC)			
3.4	Request for Technical Information (RTI)			
4	**Conduct Technical Investigations (TI)**	**Method:** Case Study **Media:** PowerPoint / Handouts **Environment:** Classroom	**70**	
4.1	Reference	Case study		
4.2	Purpose	Case study		
4.3	Occasions for conducting	Case study		
4.4	Process; 1. Terms of Reference (TOR); 2. Investigation; and 3. Reports/Recommendations.	Case study		
4.5	Agencies; 1. DRDC; 2. FTA; 3. FMFs; 4. NETE; and 5. FSRs / OEM.	Case study		
4.6	Results	Case study		
4.7	Technical Directives and Orders; 1. Restrictions; 2. SOPs; 3. UCRs; and 4. ECs.	Case study		

5. Total EO Time: Total: 2 periods as follows:

a. 2 x 50 min period – Classroom.

6. Training and Learning Aids: Nil.

7. Remarks: N/A

PO 004 – SUPPORT ENGINEERING ANALYSIS

EO 004.02 Describe Engineering Analysis Tools

1. Performance: Describe Engineering Analysis Tools.

2. Conditions:

   a. Given:

      (1) Nil;

   b. Denied:

      (1) N/A; and

   c. Environment:

      (1) Classroom.

3. Standard: The NTO shall describe Engineering Analysis Tools by:

   a. Describing Equipment Health Monitoring (EHM);

   b. Describing Reliability, Availability, Maintainability (RAM) analysis; and

   c. Describing External Agencies.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Equipment Health Monitoring (EHM)**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**75**	**A1; A3; A6; A12; A14; A15; A16; A17; A18; A22; A24; A28; A30; A33; A37; A38; A42; A48; A52; A53**
1.1	Purpose			
1.2	Types: 1. Vibrations Analysis (VA); 2. Oil and Coolant Condition Analysis Program (OCCAP); 3. Hull Surveys; and 4. Non Destructive Testing; i. X Ray; ii. Dye Penetrate; iii. Magnetic;			

Teaching Points		Instructional Strategies	Time (min)	References
	iv. Thermal Imaging; and v. Ultrasonic			
2	**Describe Reliability, Availability, Maintainability (RAM) analysis**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**75**	
2.1	Purpose/Role			
2.2	Definitions and concepts			
2.3	DGMEPM OPI			
2.4	Link to DRMIS			
3	**Describe External Agencies**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
3.1	DRDC (Dockyard Lab)			
3.2	FTA			
3.3	FSR/OEM			
3.4	FMF			
3.5	QETE			
3.6	NETE			

5. Total EO Time: Total: 4 periods as follows:

   a. 4 x 50 min period – Classroom.

6. Training and Learning Aids: Nil.

7. Remarks: N/A

PO 005 – DESCRIBE TRAINING AND PERSONNEL MANAGEMENT

EO 005.01 Describe Training of Naval Personnel

1. Performance: Describe Training of Naval Personnel.

2. Conditions:

   a. Given:

      (1) Nil;

   b. Denied:

      (1) N/A; and

   c. Environment:

      (1) Classroom.

3. Standard: The NTO shall describe the training of Naval Personnel by:

   a. Describing Personnel Policy and Procedures;

   b. Describing Organizations;

   c. Describing Career Progression; and

   d. Describing Ship's Company training.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Equipment Health Monitoring (EHM)**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**75**	**A1; A3; A6; A12; A14; A15; A16; A17; A18; A22; A24; A28; A30; A33; A37; A38; A42; A48; A52; A53**
1.1	Purpose			
1.2	Types: 5. Vibrations Analysis (VA); 6. Oil and Coolant Condition Analysis Program (OCCAP); 7. Hull Surveys; and 8. Non Destructive Testing; i. X Ray;			

Teaching Points		Instructional Strategies	Time (min)	References
	ii. Dye Penetrate; iii. Magnetic; and iv. Ultrasonic			
2	**Describe Reliability, Availability, Maintainability (RAM) analysis**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**75**	
2.1	Purpose/Role			
2.2	Definitions and concepts			
2.3	DGMEPM OPI			
2.4	Link to DRMIS			
3	**Describe External Agencies**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
3.1	DRDC (Dockyard Lab)			
3.2	FTA			
3.3	FSR/OEM			
3.4	FMF			
3.5	QETE			
3.6	NETE			

5. Total EO Time: Total: 4 periods as follows:

   a. 4 x 50 min period – Classroom.

6. Training and Learning Aids: Nil.

7. Remarks: N/A

PO 005 - DESCRIBE TRAINING AND PERSONNEL MANAGEMENT

EO 005.02 Describe Personnel Management

1. Performance: Describe Personnel Management

2. Conditions:

   a. Given:

      (1) Nil;

   b. Denied:

      (1) N/A; and

   c. Environment:

      (1) Classroom.

3. Standard: The NTO shall describe Personnel Management by:

   a. Describing Personnel Organizations;

   b. Describing Military Personnel Policies;

   c. Describing Civilian Personnel Policies.

   d. Applying Military Personnel Policies IAW the OJPR (Annex E to this document);

   e. Administering Corrective Action IAW the OJPR (Annex E to this document);

   f. Applying Career Policies IAW the OJPR (Annex E to this document);

   g. Utilizing Canadian Forces Personnel Appraisal System (CFPAS) IAW the OJPR (Annex E to this document); and

   h. Maintaining Documentation IAW the OJPR (Annex E to this document).

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Personnel Organizations**	**Method:** Lecture	**550**	**A1; A2; A6; A7; A22; A25; A26; A32; A33;**

Teaching Points		Instructional Strategies	Time (min)	References
		Media: PowerPoint **Environment:** Classroom		**A34; A37; A38; A53; A55; A58; A62; A78**
1.1	NDHQ; 1. ADM HR (CIV); 2. Chief Military Personnel (CMP); i. D MIL C; 3. Naval Staff Headquarters (NSHQ); i. DGNP; ii. Navy PERS Policy; and iii. MOC Mangers; 4. ADM (MAT) i. Branch Advisors; and ii. MOC Advisor.			
1.2	Formation Personnel: 1. PCC; 2. Transition Unit; and 3. MOC Advisors.			
1.3	Support agencies; 1. Base Personnel Selection Office (BPSO); 2. Regional Support Social Worker; 3. Formation Chaplain's Office; 4. Dispute Resolution Centre; 5. Addiction Counsellor; 6. Financial Counsellor; 7. Canadian Forces Member Assistance Program (CFMAP); 8. Ombudsman; and 9. Military Family Resource Centre (MFRC).			
2	**Describe Military Personnel Policies**	**Method:** Case Study **Media:** PowerPoint	**850**	

Teaching Points		Instructional Strategies	Time (min)	References
		Environment: Classroom		
2.1	Divisional Systems			
2.2	Divisional Issues such as: 1. Grievance; 2. Misuse of drugs; 3. Misuse of alcohol; 4. Civilian offences; 5. Harassment; 6. Indebtedness; 7. Domestic abuse; 8. Sexual Misconduct; and 9. Marital.			
2.3	Corrective Action; 1. Administrative; a. Initial Counselling (IC); b. Recorded Warning (RW); c. Counselling and Probation (C&P); and d. Administrative Review (AR). 2. Disciplinary.			
2.4	Career Policies including: 1. Occupation Transfers/reassignments; 2. Postings; 3. Promotions; and 4. Terms of Service;			
2.5	Canadian Forces Personnel Appraisal System (CFPAS or PACE)			
2.6	Documentation; 1. Unit Personnel File (UPF); 2. Monitor Mass; 3. Human Resource Management System (HRMS – PeopleSoft); and, 4. Feedback Notes; and			

Teaching Points		Instructional Strategies	Time (min)	References
3	**Describe Civilian Personnel Policies**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**200**	
3.1	Classification Reform			
3.2	Work Descriptions			
3.3	PRRs			
3.4	Unions: 1) Reps; and 2) Stewards			
3.5	Documentation; 1. Collective Agreement; 2. Personnel Files; and 3. CPAOs.			
3.6	Staffing			
3.7	Grievance Process			
3.8	Classification; 1. Competition Process; 2. Deployments/Lateral Transfers; and 3. Temporary Assignments/Acting.			
4	**Apply Military Personnel Policies;**	**Method:** OJT **Media:** N/A **Environment:** Ship		
4.1	Divisional System			
4.2	Divisional Issues such as; 1. Grievance; 2. Misuse of drugs; 3. Misuse of alcohol; 4. Civilian offences; 5. Harassment; 6. Indebtedness; 7. Domestic abuse; 8. Sexual Misconduct; and 9. Marital.			
5	**Administer Corrective Action;**	**Method:** OJT **Media:** N/A **Environment:** Ship		

Teaching Points		Instructional Strategies	Time (min)	References
5.1	Administrative; 1. Initial Counselling (IC); 2. Recorded Warning (RW); 3. Counselling and Probation (C&P); and 4. Administrative Review (AR).			
6	**Apply Career Policies including**	**Method:** OJT **Media:** N/A **Environment:** Ship		
6.1	Occupations			
6.2	Transfers/reassignments			
6.3	Postings			
6.4	Promotions			
3.5	Terms of Service			
7	**Utilize Canadian Forces Personnel Appraisal System (CFPAS)**	**Method:** OJT **Media:** N/A **Environment:** Ship		
8	**Maintain Documentation**	**Method:** OJT **Media:** N/A **Environment:** Ship		
8.1	Unit Personnel File (UPF)			
8.2	Monitor Mass			
8.3	Human Resource Management System (HRMS – Peoplesoft)			
8.4	Feedback Notes			

5. Total EO Time: Total: 32 periods as follows:

   a. 32 x 50 min period – Classroom.

6. Training and Learning Aids: Nil.

7. Remarks: N/A

PO 006 – DESCRIBE NAVAL MATERIEL MANAGEMENT

EO 006.01 Describe Naval Maintenance Policy

1. Performance: Describe Naval Maintenance Policy

2. Conditions:

   a. Given:

      (1) Nil;

   b. Denied:

      (1) N/A; and

   c. Environment:

      (1) Classroom.

3. Standard: The NTO shall describe Naval Maintenance Policy, by:

   a. Describing Naval Materiel management (NaMMs);

   b. Describing Maintenance Management Information Systems;

   c. Describing Maintenance Philosophy;

   d. Describing Maintenance levels;

   e. Describing the 5 Year Plan (Fleet Support Plan);

   f. Describing Test Equipment Calibration and Monitoring (TEMMIS);

   g. Describing Naval Engineering Manual (NEM);

   h. Describing National Defence Index of Documentation (NDID) Coding; and

   i. Describing Hierarchy of Publications.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Naval Materiel Management (NaMMs)**	**Method:** Lecture **Media:** PowerPoint	**10**	**A2; A3; A14; A15; A16; A17; A21; A23; A25; A28; A29; A30; A31; A32; A38; A39**

Teaching Points		Instructional Strategies	Time (min)	References
		Environment: Classroom		
1.1	Purpose and definition			
1.2	References			
2	**Describe Maintenance Management Information Systems (DRMIS)**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**5**	
3	**Describe Maintenance Philosophy**	**Method:** Lecture **Media:** Instructor **Environment:**	**10**	
3.1	Repair by Replacement (RxR)			
3.2	Maintenance by Exchange (MxE)			
4	**Describe Maintenance Levels**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**10**	
4.1	First Line			
4.2	Second Line			
4.3	Third Line			
5	**Describe 5 Year Plan (Fleet support plan)**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**20**	
5.1	Short Work Period (SWP)			
5.2	Extended and Docking Work Periods (EWP/DWP)			
5.3	Engineering Change Work Periods (ECWP)			
6	**Describe Test Equipment Calibration and Monitoring**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**5**	
7	**Describe Naval Engineering Manual (NEM)**	**Method:** Lecture	**10**	

Teaching Points		Instructional Strategies	Time (min)	References
		Media: PowerPoint **Environment:** Classroom		
8	**Describe National Defence Index of Documentations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**10**	
9	**Describe Hierarchy of Publications**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**20**	
9.1	NDA/QR&O's			
9.2	CFAOs/DAODs			
9.3	NaMMs			
9.4	NEM			
9.5	CFTOs			
9.6	MARCORDS			
9.7	MARLANT ORDs/MARPAC ORDs			
9.8	SSOs			
9.9	CSOs			
9.10	COTMs			
9.11	Engineering Departmental Technical Instruction			
9.12	Engineering Departmental Technical Memorandums			
9.13	Naval Engineering and Maintenance Technical Instructions (NEMTI)			

5. Total EO Time: Total: 2 periods as follows:

   a. 2 x 50 min period – Classroom.

6. Training and Learning Aids: Nil.

7. Remarks: N/A

PO 006 – DESCRIBE NAVAL MATERIEL MANAGEMENT

EO 006.02 Describe Naval Maintenance Organizations

1. Performance: Describe Naval Maintenance Organizations

2. Conditions:

   a. Given:

      (1) Nil;

   b. Denied:

      (1) N/A; and

   c. Environment:

      (1) Classroom.

3. Standard: The NTO shall describe Naval Maintenance Organizations by describing their roles and inter-relationships.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**NDHQ Organization**	**Tours**	**3 days**	**A2; A3; A14; A15; A16; A17; A21; A23; A25; A28; A29; A30; A31; A32; A38; A39**
1.1	Assistant Deputy Minister (ADM) Material; 1. Director General Maritime Engineering Program Management (DGMEPM): i. Director Major Surface Combatant (MEPM MSC); ii. Director Non-Combatant (MEPM NC); iii. Director Maritime Management Support (DMMS); iv. Naval Engineering Test Establishment (NETE); and			

Teaching Points		Instructional Strategies	Time (min)	References
	v. Quality Engineering Test Establishment (QETE). 2. Director General Major Project Delivery (Land and Sea) DGMPD (L&S)).			
1.2	Chief of Naval Staff (CRCN); 1. Director General Maritime Force Development (DGMFD); i. Director Maritime Strategy (D Mar Strat); ii. Director Maritime Requirements (Sea) (DMRS); and iii. Director Maritime Support Capability Requirements (DSCR).			
2	**Formation**	**Tours**	**200**	
2.1	Maritime Forces Atlantic (MARLANT)/Maritime Forces Pacific (MARPAC): 1. Operations Readiness and Engineering Operations; 2. Material: i. Formation Technical Authority (FTA);			
2.2	Fleet Maintenance Organization			
2.3	Fleet Diving Unit; 1. Underwater inspection/repairs			
3	**Ship's Staff**	**Tours**		
3.1	Planned Maintenance Co-ordinator (Executive Officer)			
3.2	Departmental Maintenance Co-ordinator (DMC)			
4	**Canadian Forces Supply System (CFSS)**	**Tours**		

5. Total EO Time: Tours over 5 days

6. Training and Learning Aids: Nil.

7. Remarks:

   a. Two periods of Classroom instruction, a three day tour of Naval Staff HQ / DGMEPM / DGMPD(L&S) / NETE (Ottawa/Montreal) excluding travel times, and a four period tour of FMFCS/CB (as applicable).

   b. Ottawa tour and FMF tour.

PO 006 – DESCRIBE NAVAL MATERIEL MANAGEMENT

EO 006.03 Describe Naval Maintenance Procedures

1. Performance: Describe Naval Maintenance Procedures

2. Conditions:

   a. Given:

      (1) Nil;

   b. Denied:

      (1) N/A; and

   c. Environment:

      (1) Classroom.

3. Standard: The NTO shall describe Naval Maintenance Procedures by:

   a. Describing Maintenance Administration in HMC Ships;

   b. Describing the different types of Short Work Periods (SWP);

   c. Describing Extended and Docking Work Period (EWP/DWP);

   d. Describing Support Agencies;

   e. Participating in maintenance activities IAW the OJPR (Annex E to this document); and

   f. Planning and supporting a EWP/DWP/SWP IAW the OJPR (Annex E to this document).

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Maintenance Administration in HMC Ships**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	**A2; A3; A14; A15; A16; A17; A21; A23; A25; A28; A29; A30; A31; A32; A38; A39**
1.1	Planned Maintenance; 1. Load Schedule;			

Teaching Points		Instructional Strategies	Time (min)	References
	i. Maintenance Periodicity; and 2. Performance Tests.			
1.2	Corrective Maintenance			
1.3	DRMIS introduction; 1. Purpose; 2. Notifications; i. N1 (corrective Maintenance); ii. N2 (Cost Recovery); iii. N3 (Trials); iv. N7 (TRANREQ); v. N9 (Preventive Maintenance); and vi. ND (Deviations and Waivers); 3. Relationship between Orders and Notifications.			
2	**Describe Directed Maintenance Period (DMP)**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
2.1	Process			
2.2	Milestones			
2.3	Contract			
2.4	Quality Control (QC) and Quality Assurance (QA)			
3	**Describe Docking Work Period (DWP) / Refits**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
3.1	Process			
3.2	Milestones			
3.3	Contract			
3.4	Documentation; 1. Standard Work Package; and 2. Deviation Process;			

Teaching Points		Instructional Strategies	Time (min)	References
3.5	Quality Control (QC) and Quality Assurance (QA)			
3.6	Approving Authorities			
4	**Describe Support Agencies**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
4.1	FMF			
4.2	Fleet and Formation Support; 1. FER; and 2. FTA;			
4.3	In-service Support Contract (ISSC); and			
4.4	Field Service Rep (FSR) 1. Original Equipment Manufacture (OEM); and 2. Technical Authority (LCMM) Contractor support.			
5	**Participate in the following maintenance activities**	**Method:** OJT **Media:** N/A **Environment:** Ship		
5.1	Accompany the Ship's FMF Project Leader Ship's Services Officer for one day during a DMP			
5.2	Fleet Maintenance Prioritization Meetings			
5.3	Review monthly planned maintenance with Section Heads			
5.4	Co-ordinate daily departmental maintenance activities/priorities and liaise with external agencies			
6	**Plan and support a EWP/DWP/SWP**	**Method:** OJT **Media:** N/A **Environment:** Ship		

5. Total EO Time: Total: 4 periods as follows:

   a. 3 x 50 min period – Classroom.

6. Training and Learning Aids: Nil.

7. Remarks: N/A

PO 006 - DESCRIBE NAVAL MATERIEL MANAGEMENT

EO 006.04 Describe Naval Maintenance Documentation

1. Performance: Describe Naval Maintenance Documentation

2. Conditions:

   a. Given:

      (1) Nil;

   b. Denied:

      (1) N/A; and

   c. Environment:

      (1) Classroom.

3. Standard: The NTO shall describe Naval Maintenance Documentation and Reporting, by:

   a. Explaining the purpose, completion and staffing of Naval Maintenance Documentation and Reporting; and

   b. Assisting in the Naval Maintenance Program IAW the OJPR (Annex E to this document).

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**For each of the following Maintenance Documents, explain their purpose, completion and staffing**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**150**	**A2; A3; A14; A15; A16; A17; A21; A23; A25; A28; A29; A30; A31; A32; A38; A39**
1.1	Unsatisfactory Condition Report (UCR) procedure			
1.2	Pre-installation failure (PIF)			
1.3	Supply demand			
1.4	Gas turbine running hours			
1.5	Diesel running hours			
1.6	Combat Readiness Requirement (CRR) report			

Teaching Points		Instructional Strategies	Time (min)	References
1.7	Cathodic protection report			
1.8	Periodic weight report			
1.9	AFSO report			
1.10	Fire safety report			
1.11	Signature management report			
1.12	Monthly Ammunition Expenditure report			
1.13	Monthly Maintenance Reports			
1.14	Monthly fuel report			
2	**Assisting in Naval Maintenance Program to the NTO Department Head (without references) to include the following**	**Method:** OJT **Media:** N/A **Environment:** Ship		
2.1	Planned maintenance			
2.2	Scheduling			
2.3	Maintenance Information System			
2.4	Corrective Maintenance			
2.5	Repair Facility Work Process			
2.6	Work Prioritization			
2.7	Maintenance Documentation			

5. Total EO Time: Total: 3 periods as follows:

   a. 3 x 50 min period – Classroom.

6. Training and Learning Aids: Nil.

7. Remarks: N/A

PO 006 – DESCRIBE NAVAL MATERIEL MANAGEMENT

EO 006.05 Describe Life Cycle Materiel Management (LCMM)

1. Performance: Describe Life Cycle Materiel Management (LCMM)

2. Conditions:

   a. Given:

      (1) Nil;

   b. Denied:

      (1) N/A; and

   c. Environment:

      (1) Classroom.

3. Standard: The NTO shall describe Life Cycle Materiel Management (LCMM) by:

   a. Describing Organizations (Roles and Responsibilities);

   b. Describing Life Cycle Phases;

   c. Describing Logistics Support Management;

   d. Describing Technical Data Management; and

   e. Preparing and staffing Technical Data Management Documentation IAW the OJPR (Annex E to this document).

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Organizations (Roles and Responsibilities)**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	**A2; A3; A14; A15; A16; A17; A21; A23; A25; A28; A29; A30; A31; A32; A38; A39**
1.1	DGMEPM; 1. MEPM MSC/NC (TAs); 2. DMSS (Fleet Management) 3. DMMS (Item Managers).			

Teaching Points		Instructional Strategies	Time (min)	References
1.2	Formation; 1. FMF; and 2. Ships.			
2	**Describe Life Cycle Phases**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
2.1	Conception/Definition (R&D, design etc)			
2.2	Acquisition			
2.3	In-Services Support; 1. Fitted Equipment List (FEL); 2. Equipment Support List (ESL); 3. Special tools and test equipment; 4. Maintenance routine development; 5. Spares; 6. Equipment upgrades; and 7. Technical Advisories and Bulletins.			
2.4	Disposal; 1. Procedures			
3	**Describe Logistics Support Management**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
3.1	Integrated Logistics Support (ILS); 1. Initial Procurement; and 2. Provision of spares.			
4	**Describe Technical Data Management**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
4.1	Technical Data Package (TDP)			

Teaching Points		Instructional Strategies	Time (min)	References
4.2	Methods; 1. Unsatisfactory Condition Report (UCR); 2. Engineering Change (EC); 3. Pre-Installation Failure (PIF); and 4. Operational Deficiencies (OPDEFs).			
5	**Prepare and staff the following Technical Data**	**Method:** OJT **Media:** N/A **Environment:** Ship		
5.1	Pre-Installation Failure (PIF) Report			
5.2	Unsatisfactory Condition Report (UCR)			
5.3	Engineering Change (ECP) Proposals			

5. Total EO Time: Total: 2 periods as follows:

   a. 2 x 50 min period – Classroom.

6. Training and Learning Aids: Nil.

7. Remarks:

   a. Para 7 is to be consolidated during tour of DGMEPM as identified in EO 006.02 - Describe Naval Maintenance Organizations.

PO 006 – DESCRIBE NAVAL MATERIEL MANAGEMENT

EO 006.06 Describe Naval Configuration Management

1. Performance: Describe Naval Configuration Management

2. Conditions:

   a. Given:

      (1) Nil;

   b. Denied:

      (1) N/A; and

   c. Environment:

      (1) Classroom.

3. Standard: The NTO Shall describe Naval Configuration Management by:

   a. Defining Naval Configuration Management;

   b. Describing the purpose of Naval Configuration Management;

   c. Describing the Policy of Naval Configuration Management;

   d. Describing the types of Engineering Change (EC);

   e. Describing Organizations roles and responsibility;

   f. Describing Naval Configuration Management process; and

   g. Describing EC Process.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Define Naval Configuration Management**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**10**	**A2; A3; A14; A15; A16; A17; A21; A23; A25; A28; A29; A30; A31; A32; A38; A39**

Teaching Points		Instructional Strategies	Time (min)	References
2	**Describe the purpose of Naval Configuration Management**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**10**	
3	**Describe the Policy of Naval Configuration Management for the following**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**10**	
3.1	HDW/JSS Class			
3.2	HAL Class			
3.3	VIC Class			
3.4	KIN Class			
3.4	Requirement for and EC			
4	**Describe the types of Engineering Change (EC)**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**10**	
4.1	Permanent; 1. SHIPALT; 2. SHIPMOD; 3. CANAVMOD; and 4. Software change.			
4.2	Temporary; 1. EVAL; and 2. Mission Fit.			
5	**Describe the following organizations roles and responsibility**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**20**	
5.1	Director Naval Requirements (DNR); 1. Project Directors.			
5.2	MEPM Major Surface Combantant and Non-Combatantt (MSC / NC); 1. Technical Authorties; and			

Teaching Points		Instructional Strategies	Time (min)	References
	2. System Authorities.			
5.3	Director Maritime Support Services; 1. Fleet Management.			
5.4	Fleet/Formation; 1. FER/FEO/Formation Technical Authority (FTA); and 2. FMF.			
5.5	Ships; 1. MSEO; and 2. CSEO.			
6	**Describe Naval Configuration Management process for the following**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**20**	
6.1	Audits			
6.2	ECs (from initiation to implementation)			
6.3	CIWG (Configuration Implementation Working Group)			
6.4	EVAL Semi Annual Conference			
7	**Describe EC process**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**20**	
7.1	Part 1			
7.2	Part 2			
7.3	NDHQ Project File			
7.4	DRMIS Notification			
7.5	C of C			
7.6	TDP Changes			

5. Total EO Time: Total: 2 periods as follows:

   a. 2 x 50 min period – Classroom.

6. Training and Learning Aids: Nil.

7. Remarks: N/A

PO 006 – DESCRIBE NAVAL MATERIEL MANAGEMENT

EO 006.07 Describe Engineering Quality Management

1. Performance: Describe Engineering Quality Management

2. Conditions:

   a. Given:

      (1) Nil;

   b. Denied:

      (1) N/A; and

   c. Environment:

      (1) Classroom.

3. Standard: The NTO shall describe Engineering Quality Management by:

   a. Describing the principles of Engineering Quality Management terms;

   b. Describing the roles and responsibilities of the Engineering Quality Management Organizations;

   c. Describing the Engineering Quality Management policies; and

   d. Describing the Engineering Quality Management Standards and Classifications societies.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe the principles of the following Engineering Quality Management terms**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	**A2; A3; A14; A15; A16; A17; A21; A23; A25; A28; A29; A30; A31; A32; A38; A39**
1.1	Quality Assurance (QA)			
1.2	Quality Control (QC)			
1.3	Engineering Quality Management			

Teaching Points		Instructional Strategies	Time (min)	References
2	**Describe the roles and responsibilities of the following Engineering Quality Management**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
2.1	QAWC			
2.2	QETE			
2.3	FMF			
2.4	Ships			
3	**Describe the following Engineering Quality Management policies**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
3.1	Performance Measurement Techniques			
4	**Engineering Quality Management standards and**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
4.1	Military Standards; 1. STANAG; 2. Mil Spec; 3. DODS; and 4. ANEP 77.			
4.2	Civilian Standards; 1. ISO; 2. ISA/EIA/IEEE; 3. Canadian Shipping Act (CSA); 4. ASME; 5. IMO; 6. SOLAS; 7. ASTN; and 8. ASE.			
4.3	Classification Societies; 1. Lloyds Register (LR); 2. America Bureau of Shipping (ABS); 3. Germanischer Lloyds (GL); and			

Teaching Points		Instructional Strategies	Time (min)	References
	4. Net Norske VERITAS (DNV).			

5. Total EO Time: Total: 2 periods as follows:

   a. 2 x 50 min period – Classroom.

6. Training and Learning Aids: Nil.

7. Remarks: N/A

PO 006 – DESCRIBE NAVAL MATERIEL MANAGEMENT

EO 006.8 Describe Naval Material Assurance

1. Performance: Describe Naval Material Assurance

2. Conditions:

   a. Given:

      (1) Nil;

   b. Denied:

      (1) N/A; and

   c. Environment:

      (1) Classroom.

3. Standard: The NTO Shall describe Naval Material Assurance by:

   a. Defining Naval Material Assurance

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Naval Material Assurance**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
2	**Naval Classification History**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
3	**Naval Ship Code**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**100**	
4	**Differences between commercial and naval classification;**	**Method:** Lecture	**100**	

Teaching Points		Instructional Strategies	Time (min)	References
		Media: PowerPoint **Environment:** Classroom		
5	**Benefits classification offer to a Naval Administration;**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**100**	
6	**Class notations and Classification process;**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**100**	
7	**Military aspects of Rules for Naval Ships;**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**100**	
8	**Design process and software tools; and**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**100**	
9	**Introduction to Rules for structure, military, machinery, electrical, control and statutory aspects.**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**100**	

5. Total EO Time: Total: 16 x 50 minute periods as follows:

   a. 16 x 50 min period – Classroom.

6. Training and Learning Aids: Nil.

7. Remarks: Currently being deliver by Lloyd's Registry under Introduction to Naval Classification.

PO 007 – DESCRIBE BUISNESS MANAGEMENT

EO 007.01 Describe Contract Management

1. Performance: Describe Contract Management

2. Conditions:

   a. Given:

      (1) Nil;

   b. Denied:

      (1) N/A; and

   c. Environment:

      (1) Classroom.

3. Standard: The NTO shall describe Contract Management by:

   a. Describing the roles and responsibilities of the following organizations in the contracting process;

   b. Describing contracting policy;

   c. Describing the contracting process;

   d. Describing contract types;

   e. Describing contracting indiscretions; and

   f. Describing conflict of interest guidelines.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe the roles and responsibilities of the following organizations in the contracting process**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	**A2; A6**
1.1	PSPC			
1.2	Formation Logistics			

Teaching Points		Instructional Strategies	Time (min)	References
1.3	Ships			
2	**Describe contracting policy**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**10**	
2.1	1) Elements of a Legal Contract; 2) Key Principles; i. Best Value; ii. Competition; iii. Fairness; iv. Operational Requirements; v. Socio-economic Considerations; and vi. Prudence and Probity; 3) Contracting Authority; 4) Spending Authority; 5) Payment Authority; 6) Trade Agreements; and 7) Statutes.			
3	**Describe the contracting process**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
3.1	Initiation; 1) Identifying Requirements; 2) Key Considerations; and 3) Sourcing the Requirement.			
3.2	Negotiation; 1) Bid Solicitation; 2) Bid Submission; 3) Bid Evaluation; 4) Contract Award; and 5) Disclosures.			
3.3	Monitor 1) Contract Management.			

Teaching Points		Instructional Strategies	Time (min)	References
3.4	Review; 1) Confirmation of Performance and Price; 2) File Management; 3) Distribution Accounts: i. Sub Unit Accounts;			
4	**Describe contract types**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
4.1	Full			
4.2	Competitive			
4.3	Non-Competitive; 1) Sole Sourcing; and 2) Standing Offers.			
4.4	Abbreviated			
4.5	Credit Card; 1) Policy; 2) Purchase/Contracting Limits; 3) Goods and Services Lists; and 4) Locally Controlled Purchase Accounts			
4.6	Internal; 1) Service Level Agreements ; and 2) Memorandums of Understanding			
5	**Describe contracting indiscretions**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
5.1	Full			
5.2	Competitive			
5.3	Non-Competitive; 1) Sole Sourcing; and 2) Supply Arrangements and Standing Offers.			

Teaching Points		Instructional Strategies	Time (min)	References
5.4	Abbreviated			
5.5	Credit Card; 1) Policy; 2) Purchase/Contracting Limits; 3) Goods and Services Lists; and 4) Locally Controlled Purchase Accounts			
5.6	Internal; 1) Service Level Agreements ; and 2) Memorandums of Understanding			
6	**Describe contracting indiscretions**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
6.1	Contract Splitting			
6.2	Retroactive Contracting			
6.3	Employer / Employee Relationship			
6.4	Pay rolling			
6.5	Non-Rotational Sourcing			
6.6	Non-Competitive Contracts			
7	**Describe conflict of interest guidelines**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	

5. Total EO Time: Total: 4 x 50 minute periods as follows:

   a. 4 x 50 min period – Classroom.

6. Training and Learning Aids: Nil.

7. Remarks: N/A

PO 007 – DESCRIBE BUSINESS MANAGEMENT

EO 007.02 Describe Capability Planning

1. Performance: Describe Capability Planning.

2. Conditions:

   a. Given:

      (1) Nil;

   b. Denied:

      (1) N/A; and

   c. Environment:

      (1) Classroom.

3. Standard: The NTO shall describe Capability Planning by:

   a. Describing the capability planning roles and responsibilities;

   b. Describing the following capability planning principles; and

   c. Describing strategic planning and reporting process.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe the capability planning roles and responsibilities for the following organizations**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	**A2; A6**
1.1	VCDS			
1.2	CNS			
1.3	Formations			
1.4	Units/Ships			
2	**Describe the following capability planning principles**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	

Teaching Points		Instructional Strategies	Time (min)	References
2.1	Strategic Visioning			
2.2	Capability Based Planning			
2.3	Resource Prioritization/Management			
2.4	Performance Management			
2.5	Risk Management			
3	**Describe strategic planning and reporting process**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**150**	
3.1	Reporting Levels			
3.2	Guidance Documents such as the following: 1. Planning Reporting & Accounting Structure (PRAS); 2. Strategy 2020; 3. White Paper 1994; 4. Defence Planning Guide; 5. Strategic Planning Guide (CRCN); and 6. Formation Planning Guidance.			
3.3	Formation Responsibilities			
3.4	Unit Responsibilities; 1. Strategic Planning i. Mission Statement; ii. Vision Statement; iii. Organizational Core Values; iv. Past Practices Review; v. Identification of Goals/Priorities; vi. Creating the Plan; vii. Measuring Progress; 2. Business Planning (BP) i. BP Process; 3. Fiscal Year Timings; 4. Budget Estimations; 5. Revenue Generation; 6. Cost Recovery 7. Expenditure Trend Analysis 8. Creating the Plan; 9. Amendments;			

Teaching Points		Instructional Strategies	Time (min)	References
	i. Activity Based Costing (ABC); and 10. Definition; 11. Implementation; 12. Modelling; 13. Software; 14. Performance Measurement. i. Performance Measurement / Performance Reporting; 15. Balanced Scorecard; 16. Budget Management; i. Forecasting; ii. Tracking / Reporting; iii. Accrual Accounting; iv. O&M to Capital Conversions v. Progressive Surplus Taxation; vi. FY End Process; vii. Payable at Year End (PAYE).			
3.5	Definitions; 1. Responsibility Centres; 2. Cost Centres; 3. Financial Codes; 4. Forecast/Committed/Expended; 5. O&M; and 6. Capital.			
3.6	Management Aids; 1. DRMIS (Materiel and Financial); 2. AMEX Reports.			
3.7	Business Case Analysis; 1. Alternate Service Delivery (ASD); 2. Business Process Re-engineering.			

5. Total EO Time: Total: 4 periods as follows:

   a. 4 x 50 min period – Classroom.

6. Training and Learning Aids: Nil.

7. Remarks: N/A

PO 008 – MANAGE PROJECTS

EO 008.01 Describe Project Management Principles

1. Performance: Describe Project Management Principles.

2. Conditions:

   a. Given:

      (1) Nil;

   b. Denied:

      (1) N/A; and

   c. Environment:

      (1) Classroom.

3. Standard: The NTO shall describe project management principles by:

   a. Describing Project Management terminology;

   b. Describing the Project Process;

   c. Describing project Stakeholders including the roles and importance of each;

   d. Describing organizational influences on Project Management (description, advantages/disadvantages of each Project Manager's influence in each);

   e. Describing Project Management Requirements(Description and key activities in each; links and interactions to/with each other); and

   f. Describing Project Management Methods.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe Project Management terminology**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**100**	**A2; A7; A14; A15; A38; A42**

Teaching Points		Instructional Strategies	Time (min)	References
1.1	Project			
1.2	Project Management			
2	**Describe the Project Process**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**200**	
2.1	Initiating			
2.2	Planning			
2.3	Executing			
2.3	Monitoring and Controlling			
2.4	Closing			
3	**Describe project Stakeholders including the roles and importance of each**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**200**	
3.1	Project Manager			
3.2	Customer			
3.3	Performing organization			
3.4	Team members			
3.5	Sponsor			
4	**Describe organizational influences on Project Management (description, advantages / disadvantages of each Project Manager's influence in each)**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**200**	
4.1	Functional organization			
4.2	Project organizations			
4.3	Weak matrix organization			
4.4	Strong matrix organization			
4.5	Composite organization			
5	**Describe Project Management links and interactions to/with each other)**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**400**	
5.1	Project Integration Management			
5.2	Project Scope Management			
5.3	Project Time Management			

Teaching Points		Instructional Strategies	Time (min)	References
5.4	Project Cost Management			
5.5	Project Quality Management			
5.6	Project Human Resource Management			
5.7	Project Communications Management			
5.8	Project Risk Management			
5.9	Project Procurement Management			
6	**Describe Project Management Methods**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	500	
6.1	Project Scope Management; 1. Project Charter - contents, use, importance of; 2. The Project Plan - contents, use, importance of; 3. Project kick-off - importance of; 4. Work Breakdown Structure - emphasis on central role of WBS and link to other processes; 5. Scope Change Control; 6. Senior Management buy-in; and 7. Project Change Control;			
6.2	Project Time Management; 1. Activity Sequencing – network diagrams; types; 2. Activity Duration Estimating – techniques; 3. Schedule Development: i. Constraints, leads/lags; ii. CPM, GERT, PERT; iii. Bar charts, milestones charts; iv. Crashing fast tracking;			

Teaching Points		Instructional Strategies	Time (min)	References
	v. Monte Carlo simulation; vi. Resource levelling; and vii. Software tools; 4. Schedule Control; 5. Schedule baselines; 6. Change control; and 7. Performance measurement – Earned Value Management (EVM).			
6.3	Project Cost Management; 1. Cost Estimating - methods; analogous, bottom-up, parametric modeling; 2. Cost Baseline; and 3. Cost Control - Earned Value Management (EVM).			
6.4	Project Quality Management; 1. Quality Assurance/Control-purpose, definitions, differences; 2. Standards; 3. Control charts; 4. Pareto analysis; 5. Sampling; and 6. Inspections.			
6.5	Project Human Resource Management; 1. Staffing plan; 2. Team roles and responsibilities; 3. Team training; 4. Co-location; 5. Rewards and recognition; and 6. Team close out.			
6.6	Project Risk Management; 1. Risk Identification; 2. Risk Analysis; 3. Response Planning; and 4. Risk Control.			

Teaching Points		Instructional Strategies	Time (min)	References
6.7	Project Procurement Management; 1. Solicitation - statements of work, proposals; 2. Source Selection - evaluation criteria; and 3. Contract Administration.			
6.8	Project Communication Management; 1. Information Distribution; i. Internal; and ii. External; 2. Project Performance Reporting (How much? How often?); 3. Performance reviews, progress reports; 4. Variance analysis; 5. Earned Value analysis; and 6. Trend analysis.			

5. Total EO Time: Total: 32 periods as follows:

   a. 32 x 50 min period – Classroom.

6. Training and Learning Aids: Nil.

7. Remarks: N/A

PO 008 – MANAGE PROJECTS

EO 008.02 Describe the Defence Services Program (DSP)

1. Performance: Describe the Defence Services Program (DSP)

2. Conditions:

   a. Given:

      (1) Nil;

   b. Denied:

      (1) N/A; and

   c. Environment:

      (1) Classroom.

3. Standard: The NTO shall describe the Defence Services Program by:

   a. Describe the DSP Overview including definitions of the Capability programs;

   b. Describe the management of the DSP including definitions and a brief discussion;

   c. Describing project Stakeholders including the roles and importance of each;

   d. Describe Strategic Level Committee Structure;

   e. Describe Defence Management System (DMS) project Management Considerations; and

   f. Describe Industry and International Co-operation.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe the DSP Overview including definitions of the five (5) Capability programs**	**Method:** Lecture **Media:** PowerPoint	**50**	**A2; A7; A14; A15; A38; A42**

Teaching Points		Instructional Strategies	Time (min)	References
		Environment: Classroom		
1.1	Command and Control			
1.2	Conduct Operations			
1.3	Sustain Forces			
1.4	Generate Forces			
1.5	Corporate Policy and Strategy			
2	**Describe the management of the DSP including definitions and brief discussion of the following**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
2.1	Expenditure Management Program, and the Planning, Reporting and Accountability			
2.2	Parliamentary Votes for funding; 1. vote 1 – Personnel, Operations and Maintenance; 2. Vote 5 – Capital; 3. Vote 10 – Grants and Contributions; and 4. Statutory;			
3	**Describe Strategic Level Committee Structure**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	
4	**Describe Defence Management System (DMS)**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
4.1	DMS Project phases			
4.2	DMS Project approval documents			
4.3	DMS Project personnel (roles and responsibilities, place in organizations, etc.); 1. Senior Review Board; 2. Project Leader; 3. Project Director; and 4. Project Manager.			

Teaching Points		Instructional Strategies	Time (min)	References
4.4	Operational test and evaluation group; 1. Operational Test Director.			
5	**Describe Industry and International Co-operation**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**25**	

5. Total EO Time: Total: 4 periods as follows:

   a. 4 x 50 min period – Classroom.

6. Training and Learning Aids: Nil.

7. Remarks: N/A

PO 008 – MANAGE PROJECTS

EO 008.03 Conduct Engineering Design and Integration

1. Performance: Conduct Engineering Design and Integration

2. Conditions:

   a. Given:

      (1) Nil;

   b. Denied:

      (1) N/A; and

   c. Environment:

      (1) Classroom.

3. Standard: The NTO shall conduct Engineering Design and Integration by:

   a. Describing design process (Design Spiral);

   b. Describing Options analysis;

   c. Describing Detailed design;

   d. Describing Project Documents; and

   e. Completing Project and Written Report.

4. Teaching Points:

Teaching Points		Instructional Strategies	Time (min)	References
1	**Describe design process (Design Spiral)**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**100**	**A2; A7; A14; A15; A38; A42**
1.1	Problem definition/scope			
1.2	Problem analysis			
1.3	Statement of Requirements			
1.4	Conceptual design			
1.5	Options analysis			

Teaching Points		Instructional Strategies	Time (min)	References
1.6	Selection			
1.7	Detailed design			
1.8	Manufacture			
1.9	Installation			
1.10	Trials			
1.11	In-Service support			
2	**Describe Options analysis**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**100**	
2.1	Weight Factors			
2.2	Decision Matrices			
3	**Describe Detailed design**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**150**	
3.1	Design factors; 1. Weight; 2. Cost; 3. Size; 4. Reliability; 5. Redundancy; 6. Signature; 7. Material Compatibility; 8. Risk Management; 9. Ergonomics; 10. COTS vs. MILSPEC; 11. Paint and Preservation; 12. Constraints; and 13. Mounting.			
3.2	System Integration factors; 1. Environmental i. HVAC; and ii. Cooling; 2. HP/LP Air; 3. Component Placement; 4. Mounting Arrangement; 5. Electrical; 6. Weapons; 7. Sensors; 8. Navigation system; and			

Teaching Points		Instructional Strategies	Time (min)	References
	9. CCS.			
3.3	Considerations; 1. Stability; i. Weight Control 2. Tests and Trials; 3. Regulations and Legislation; i. Environmental impact; and ii. Safety; 4. Reliability, Availability and Maintainability (RAM); 5. Integrated Logistical Support (ILS) i. Spares; ii. Training; iii. Documentation; and iv. Personnel Requirements; 6. Scheduling; and 7. Habitability.			
4	**Describe Project Documents**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	
4.1	Statement of Capability Deficiency (SOCD)			
4.2	Statement of Requirements (SOR)			
4.3	Technical Statement of Requirements (TSOR)			
4.4	Statement of Work (SOW)			
4.5	Request of Proposals (RFP)			
4.6	Technical drawings			
5	**Complete Project and Written Report**	**Method:** Lecture **Media:** PowerPoint **Environment:** Classroom	**50**	

5. Total EO Time: Total: 9 periods as follows:

a. 9 x 50 min period – Classroom.

6. Training and Learning Aids: Nil.

7. Remarks: N/A

QUALIFICATION STANDARD ANNEXES

ANNEX A – MAIN REFERENCES

1. This annex lists all references used in this QS and TP.

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A		**Official CAF Publications and Directives that relate to this Qualification**
A01		A-AD-121-F01/JX-000 Manual of Abbreviations
A02		A-AE-219-00VAG-001 Organization , Concept and Policy
A03		A-QA-193-001/AG-001 Manual of DND Quality Assurance Management, Vol. I
A04		A-LM-501-001/AG-003 Code Control Guide
A05		A-LM-502-001 /FP-001 Field Supply Operation Transaction Guide
A06		A-PD-121-C01/FP-000 Staff Procedures and Military Writing
A07		A-PD-131-001/PT-001 Leadership Manual
A08		A-SJ-100-001/AS-001 Security Orders for DND, Vol. I C-01-000-000/AX-000 DID
A09		C-01-000-102/AG-000 DID Coding System
A10		C-01-007-005/DA-001 NDID Coding Chart
A11		C-02-015-001/AG-000 Instructions for Completing CF 777 Unsatisfactory
A12		Condition Report
A13		C-02-040-009/AG-001 General Safety Manual
A14		C-03-005-012/AM-001 Naval Materiel Management System
A15		C-03-005-033/AA-001 Naval Engineering Manual
A16		C-03-015-024/AM-000 Maritime Pre-Installation Failure Reporting
A17		A-GG-040-006/AG-002 Accident. Incident, Defect & Malfunctions
A18		C-09-153-001/TS-003 Ammunition Safety Manual, Vol. 3
A19		C-09-215-001/TX-000 Ammunition Restrictions
A20		C-55-040-001/TS-001 Radiation Hazard
A21		TI M.000-000/AX-000 Instructions for Completing an ESL Change Notice
A22		Canadian Forces Administrative Orders (CFAOs)
A23		CF 1305 Maintenance Action Form Log
A24		CF 411 - CF 419 Ammunition Log
A25		CFCD 102 (Current Issue)
A26		CFCD 109

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A27		CF Medical Orders (Relevant Safety Extracts)
A28		Current Maintenance Administration Bulletins (MABs)
A29		Engineering and Maintenance Readiness Inspection Checklist for Ships
A30		Engineering Change Proposal (CP Completion Guide
A31		Equipment Support List (ESL)
A32		Fleet SOPs
A33		Formation Orders
A34		Guide to Divisional System
A35		Harbour Control Instructions
A36		Maritime Command Ammunition Allocation List (MCAAL)
A37		Maritime Command Orders (MARCORDs) Vol. 1. 2 & 3A
A38		Maritime Command Ships Standing Orders (SSOs)
A39		Material Authorization Change Request (MACR) Guide
A40		Material Authorization Ships Table (MAST) Manual
A41		Missile Condition Log
A42		Naval Technical Bulletins (as required)
A43	B-0G-282-000/FP-000	Shipborne Helicopter Operating Procedures (SHOPs)
A44	C-01-007-005/DA-001	DID Coding Chart (duplicate to be adjusted)
A45		Ammunition Directives
A46		MARLANT/MARPAC Ammunition Allocation List Part 1 (MAAL/MPAAL)
A47		Identification of Ammunition
A48		Ammunition Logistic Data
A49		Captain's Ship Book
A50		Torpedo Log book
A51		Harpoon Log Book
A52		DND Explosive Safety Program
A53		OR&Os
A54	A-PM-521-001/AG-K01	Resource Manager Report (REMAR)
A55		Sea Training Guides
A56		Towed Body Log
A57		WHMIS Instruction Guide
A58		General Safety Policy Manuals
A59		Emergency Response Guidebook
A60	A-LM-187-004/JS-001	| Storage and Handling of Hazardous Materials
A61		NIOSH Pocket Guide to Chemical Hazards

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A62		SEMS
A63		Ships Standing Orders - HAZMAT Spill Response
A64		Canadian Environmental Protection Act
A65		Canadian Environmental Assessment Act
A66		Fisheries Act
A67		Transportation of Dangerous Goods Act
A68		Hazardous Products Act
A69		Arctic Waters Pollution Prevention Act
A70		Territorial Lands Act
A71		Pest Control Products Act
A72		Canada Shipping Act
A73	DAOD 4003-0	Environmental Protection & Stewardship
A74	DAOD 4003-1	Hazardous Material Management
A75		ED 4003-1/99 Spill Reporting
A76		MARCORD 4-12 Policy Directive - Environmental Program
A77		MARCORD 66-05 Hazardous Material Management
A78		Respiratory Protection Policy
A79	C-21-132-000/MS-001	Meteorological System
A80	C-21-134-003/MS-001	AN/WQQ-502(V)3 Bathythermograph Mk8AR-1662
A81	C-21-154-000/MS-001	S/D Converter Junction Box
A82	C-46-495-000/MS-001	AN/UH-504 Recorder/Reproducer Set, Signal Data (Shipboard Mass Storage System)
A83	C-46-498-000/MS-001	AN/UYK-507(V) Data Processing Set
A84	C-51-406-000/MS-001	AN/SRR-504(V) Receiving Set Radio WK 2182 C
A85	C-51-428-000/MS-000	AN/SRR-503(V) Distress Receiver
A86	C-51-501-000/MS-001	Exterior Communications System, Technical Manual
A87	C-51-528-000/MS-001	OE-82C/SC-1(V), Antenna Group, Operation and Maintenance Manual
A88	C-51-529-001/MS-003	ON-143(V)/USQ Interconnect Group, Operation and Maintenance
A89	C-51-530-000/MS-001	AN/SSR-1A Receiving Set Satellite, Operation and Maintenance
A90	C-51-530-000/MB-001	AN/SSR-1A Receiving Set Satellite, Organizational Maintenance Instructions
A91	C-51-559-000/MF-001	RT-5073(V)1/MSC-3, Communication Set Satellite, Operation and Maintenance, Vol. 1
A92	C-51-561-000/MS-001	R-5113/SR, UHF Receiver, Operation and Maintenance
A93	C-51-568-000/MS-001	RF-590 Receiver, Operation and Maintenance Manual
A94		C-51-574-000/MS-001 UHF LOS Radio Communications System Manual

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A95		C-51-575-000/MS-001 Communication Control and Monitoring System, Book 1
A96		C-51-576-000/MS-001 Ancillary Radio Equipment System Manual
A97		C-51-584-000/MS-001 HF Emergency Communication System, Technical Manual
A98		C-51-586-000/MS-001 UHF Satellite Radio Communications System Manual
A99		C-51-598-000/MS-001 OA-5180/S, Wideband Audio Switching Group, Diagrams and Parts Lists, Vol. 2
A100		C-51-637-000/MS-001 Distress Monitor Panel LS-5026/SRC
A101		C-51-639-000/MS-001 BZ-5021/U, Operator Alarm Panel, Instruction Manual
A102		C-51-809-000/MB-001 NERA Saturn BM Marine MK2 - Class 2 - Operator's Manual
A103		C-52-289-001/MS-001 AN/SSC-505(V) SHINCOM, Description and Maintenance Manual, Vol. 1
A104		C-52-289-001/MB-001 AN/SSC-505(V) SHINCOM, Description an Maintenance Manual, Vol. 4
A105		C-52-370-000/MS-001 Entertainment, Training and Briefing System, Technical Manual
A106		C-53-146-000/MN-001 AN/VRC-512, RT-524/VRC, RT-246/VRC, RT-246A/VRC, RT-524A/VRC Receiver / Transmitter Radio Set, Maintenance Manuals (2 Books)
A107		C-53-168-000/MS-001 RF-601A Antenna Coupler Group, Instruction Manual.
A108		C-53-279-000/MS-002 AN/VRC-49 Receiver/Transmitter Radio Set Operator Manual
A109		C-53-279-000/MS-003 AN/VRC-512 Receiver/Transmitter Radio Set, Operation Manual
A110		C-51-670-000/MS-001 CD-5021/UR Transmit System Control Unit, Instruction Manual
A111		C-57-716-000/MS-001 AN/URN-25 TACAN Navigation Set, Vol. 1
A112		C-57-716-000/MS-002 AN/URN-502 TACAN Navigation Set, Vol. 2
A113		C-57-716-000/MS-003 AN/URN-25 TACAN Navigation Set, Vol. 3
A114		C-57-546-000/MS-001 AN/URN-25 TACAN Navigation Antenna Group OE-273(V) and OE-273A(V). Vol. 1

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A115		C-57-546-000/MS-002 AN/URN-25 TACAN Navigation Antenna Group OE-273(V) and OE-273A(V). Vol. 2
A116		C-57-611-002/MS-001 AN/SQN-502(V) Speed Log Doppler Sonar System, Operation Manual
A117		C-57-611-002/MS-002 AN/SQN-502 Speed Log Doppler Sonar System, Service Manual
A118		C-57-624-NOO/MS-000 AN/SRN-502 Omega II Navigation Set, Operation Manual
A119		C-57-624-000/MS-001 AN/SRN-502 Omega II Navigation Set, Service Manual
A120		Not Allocated
A121		C-57-625-000/MS-001 AN/SQN-501 Sounding Set Sonar, Vol. 1
A122		C-57-625-000/MS-002 AN/SQN-501 Sounding Set Sonar, Vol. 2
A123		C-57-625-000/MY-001 AN/SQN-501 Sounding Set Sonar, Vol. 3
A124		C-57-633-A00/MS-001 AN/SYN-502 Inertial Navigation Remote Display
A125		C-57-634-COO/MS-001 CS-415 Inertial Navigation System Alarm Unit, Operation and Maintenance
A126		C-57-705-000/MS-001 AN/SSN-502 Navigation Switchboard, Operation and Maintenance, Vol. 1
A127		C-57-794-000/MS-001 Operation and Service Manual Sins-HALIFAX Ring Laser Gyro Navigator (RLGN)
A128		C-58-251-000/MB-001 AN/SLQ-501 B CANEWS, ESM Set, System Operation
A129		C-58-251-000/MS-Z01 AN/SLQ-501 B CANEWS,ESM Set, Supplement
A130		C-58 252-000/MS-001 AN/SLA-10B Blanker Video Mixer, Operation and Maintenance
A131		C-58-258-000/MS-003 AN/SLQ-502 CM Offboard System (SHIELD II),Vol. 1
A132		C-58-258-000/MS-004 AN/SLQ-502 ECM Offboard System (SHIELD I1),Vol. 2
A133		C-58-259-000/MD-001 AN/SLQ-503 RAMSES ESM Set, Theory of Operation, Vol. 1, Books 1 through 5
A134		C-58-260-CCO/MQ-001 AN/SRD-502 Surveillance Receiver, Operation and Programming, Vol. 1
A135		C-58-260-CCO/MR-001 AN/SRD-502 Surveillance Receiver, Test and Adiustments, Vol. 2

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A136		C-58-260-CCO/ML-001 AN/SRD-502 Surveillance Receiver, Troubleshooting, Vol. 3
A137		C-58-260-CCO/ML-002 AN/SRD-502 Surveillance Receiver, Troubleshooting, Vol. 4
A138		C-58-260-CCO/MB-001 AN/SD 502 Surveillance Receiver, Operator's Handbook. Vol. 5
A139		C-58-260-000/MS-001 AN/SRD-502 Direction Finder Set, Vol. 1 (2 Books)
A140		C_58-260-CDO/MS-001 AN/SRD-502 Plasma Display
A141		C-59-678-000/MS-001 AN/SPX-501(V)3 IFF Mk XII System, Operation and Maintenance
A142		C-59-723-000/MB-001 AN/SPS-49(V) Radar System, Operation, Scheduled Maintenance and General Information, Vol. 1
A143		C-59-723-000/D-001 AN/SPS-49(V) Radar System, Functional Descriptions, Vol. 2
A144		C-59-723-000/MT-001 AN/SPS-49(V) Radar System, Troubleshooting Procedures, Vol. 3
A145		C-59-723-000/MT-002 AN/SPS-49(V) Radar System, Troubleshooting Logic and Flow Diagrams, Vol. 4
A146		C-59-723-000/MT-003 AN/SPS-49(V) Radar System, Troubleshooting Logic and Flow Diagrams, Vol. 5
A147		C-59-723-000/DS-001 AN/SPS-49(V) Radar System, Troubleshooting Schematics, Vol. 6
A148		C-59-724-000/MS-001 AN/SPS-505 Radar System, Operation, Description and Block Diagrams
A149		C-59-724-DOO/MS-001 AN/SPS-505 Radar System, Cooling Unit/Dry Air Unit
A150		C-59-777-000/ML-001 AN/SPS-504 Radar System, Service Manual and Addendum. Vol. 1 (2 Books)
A151		C-59-777-000/MB-001 AN/SPS-504 Radar System, Operation Manual and Addendum. Vol. 2 (2 Books)
A152		C-59-778-XXX/XX-XXX Radiation Control Unit
A153		C-59-791-000/MS-001 Interface Unit for Radiation Control Unit
A154		C-59-765-001/MS-001 AN/UYS-503(V) Acoustic Signal Processor, Operation and Maintenance Manual, Vol. 1
A155		C-59-667-000/MS-003 Shipboard Sonobuoy Receiver Assembly, Operation and Maintenance Manual

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A156		C-59-766-000/MB-001 AN/USH-33(V) Recorder Reproducer, Operator's Manual, Vol. 1
A157		C-59-844-000/MS-001 Operation, Installation and Servicing Radar SetAN/SPS-506 Type 1007
A158		C-60-172-A00/MS-001 Radar Distribution Unit, (RDU) SensorSwitchboard, SA-5148/USQ-505 (2 Books)
A159		C-60-172-B00/MS-001 Video Switching Unit (VSU), Sensor Switchboard,SA-5151/USQ-505 (2 Books)
A160		C-160-187-000/MS-001 Weapons Status Panel, Operation and Maintenance
A161		C-60-186-A00/MS-001 AN/SSQ-532 Weapons Veto Panel, Operation and Maintenance
A162		C-60-186-CO0/MS-001 AN/SSQ-532 Weapons Veto Panel, Remote Switching Box
A163		C-69-700-000/MS-000 AN/QC-501(V) Communication Set Sonar, Underwater Telephone
A164		C-69-713-000/MS-001 AN/SLQ-25 Torpedo Countermeasures Transmitting Set, Operation and Maintenance Instructions
A165		C-69-713-MOO/MS-001 AN/SLQ-25 Control and Interface
A166		C-69-725-000/MS-001 Torpedo Launcher Control (TLC), Instruction Manual
A167		C-69-745-000/MS-001 Description, Operation, Maintenance, Towed Array Handling and Stowage Group OK-410(V)1/SQR, Vol. 1, Book 1
A168		C-69-748-000/MS-001 Maintenance Manual CANTASS Ship Electronic Subsvstem Book 1 of 2
A169		C-700-266-000/D-001 AN/SPG-503 STIR Fire Control System, Part 1, System Description
A170		C-70-266-000/MD-Z01 AN/SPG-503 STIR Fire Control System, Part 1 Supplement, System Description Supplement
A171		C-70-266-000/MS-001 AN/SPG-503 STIR Fire Control System, Gun Missile Fire Control System
A172		C-70-267-000/MD-001 Guided Missile Vertical Launching System Mk48 MOD 0, Description Manual
A173		C-70-268-000/MD-001 AN/SWG-501 Harpoon Missile System, Description and Operation, Vol. 1
A174		C-70-270-000/MB-001 57mm Gun System, Operation Manual, Vol. 1

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A175		C-70-270-000/D-001 57 mm Gun System, Description Manual, Vol. 2 (2 books)
A176		C-70-270-000/DS-001 57mm Gun System, Diagrams, Vol. 3
A177		C-70-312-000/MD-001 Transmitter Group Mk73 MOD 1, Description Manual
A178		C-70-321-000/MS-001 CIWS Introduction, Vol. 1
A179		C-70-321-000/MS-002 CIS Description, Vol. 2
A180		C-70-321-000/MB-001 CIWS Operations, Vol. 4
A181		C-70-321-000/MS-Z01 CIS Supplemental Data for Volumes 1 to 10
A182		C-70-321-000/MS-004 CIS Peculiar Support Equipment, Vol. 1
A183		C-70-321-000/MS-007 CIWS Peculiar Support Equipment, Vol. 4
A184		0967-LP-614-7010 ON-143 (V) Interconnecting Group
A185		110181-0043-01A Falcon Series HF Communication System
A186		110208-50038 RF-382 Fast Tune Auto-coupler
A187		1125-2388-045 51M-81 UHF 243MHZ Guard Receiver
A188		9853-355-000/MX-000 AN/PRC-515 Radio Set
A189		C-21-134-002/MF002 Bathy THERMOGRAPH MK8
A190		C-39-101-000/MS-001 Gyro Stabilized Horizon Reference System
A191		C-39-115-B00/MS-000 Instruction Book - Synchro Signal Re Transmission Unit
A192		C-39-115-A00/MJ-000 Instruction Manual, VG-204F Displacement Jet Gyro
A193		C-41-L00-014 AMD NULKA Maintenance Manual Vol. 1
A194		C-46-000-000/AX-000 Data Processing Equipment Index
A195		C-46-250-000/MS-001 AN/UC-503 (V) Bootstrap
A196		C-46-254-000/MS-001 AN/USH-504 (V) Q2600 Magnetic Tape Unit Operation and Maintenance Vol. 1
A197		C-46-337-000/MS-001 AN/UYK-505 (V) Data Processing Set Vol. 1
A198		C-46-380-000/DC-001 Bus Interface Set (NODE), AN/UC-503 (V) Vol. 2

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A199		C-46-380-000/DS-001 AN/UYK-502 (V) Data Processing Set Technical Manual
A200		C-46-421-000/MS-001 Operation and Maintenance Manual for SHINPADS Standard Display AN/UYQ-501 (V)
A201		C-46-448-000/DC-001 AN/UYC-504 (V) Data Processing/Bus Interface
A202		C-46-448-000/MS-001 AN/UC-504 (V) Data Processing/Bus Interface
A203		C-46-449-000/MS-000 AN/UC-504 (V) Technical Manual BAM
A204		C-46-449-000/MS-001 AN/UYC-504 (V) Technical Manual
A205		C-46-450-000/MS-001 Bus Interface Set NODE Vol. 1 Operation and Maintenance Manual with Parts List
A206		C-46-450-000/MS-Z01 AN/UYC-503 (V) Technical Manual
A207		C-46-484-002/MS-001 HSP-3609 High Speed Printer Operating and Service Manual M365-90
A208		C-46-497-000/MS-001 Data Processing and Bus Interface
A209		C-46-498-000/MS-001 AN/UYK-507 Operation and Maintenance Manual
A210		C-46-498-000/MS-Z01 AN/UYK-507 Remote Terminal User Manual
A211		C-50-251-A00/MJ-000 618M-3/3A Collins VHF Communications Transceiver
A212		C-50-251-A02/MS-000 Collins 303N Controls (5XX Series)
A213		C-51-000-000/AX-000 Naval External Communication Equipment Index
A214		C-51-007-001/MS-000 Cesium Beam Frequency Standard Hewlett Packard 5061A
A215		AMC 8/12/BC Equipment Manual Antenna Distributor
A216		C-46-495-000/MS-001 Recorder/reproducer set, Signal Data AN/UH-504
A217		C-50-254-000/MS-001 Control, Frequency, Selector C-5556/ARC-164(V)
A218		C-50-254-000/MY-000 Illustrated parts breakdown Radio Set AN/ARC-164(V)
A219		C-51-339-000/NY-001 Radio Antennas (Whip and Wire)

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A220		C-51-339-000/NY-002 Communication Antenna Group - UHF. VHF
A221		C-51-339-K00/MS-001 Ultra High Frequency Antenna AS-5042/URC
A222		C-51-050-001/MD-001 Handbook for Communications system
A223		C-51-501-000/MS-001 External Naval Communications System
A224		C-51-528-000/MS-001 Antenna Group OE-82C/WSC-1(V)
A225		C-51-528-000/MS-002 Antenna Group OE-82C/WSC-1(V)
A226		C-51-304-000/MS-000 BZ-5007/U Alarm - monitor
A227		C-51-529-000/MS-003 Interconnecting Group ON-143(V)6/USQ
A228		C-51-407-000/MS-001 Sealand 66 Maintenance Manual
A229		C-51-474-000/MS-000 AS-5104/SRC UH Polemast Antenna Assembly
A230		C-51-475-000/MS-001 OE-5012/SC Serial Control Group Interface Group
A231		C-51-475-G00/MS-001 CU-5121/SC Antenna Control Interface Module
A232		C-51-521-000/MS-000 CV5144 U Converter, Patch Panel Technical Manual
A233		C-51-529-001/MS-Z01 ON-143 (V) 6/USQ Interconnecting Group Technical Supplement 2
A234		C-51-534-000/MS-001 R-5099/U Radio Receiver Technical Manual
A235		C-51-552-000/MS-001 Technical Manual Model 1280R (MS-5062/UGC)
A236		C-51-658-000/MS-001 TD-1271 B/U Multiplexer
A237		C-51-573-000/MS-001 Operation and Maintenance Instruction with Parts Identification List, Book 1, Radio Set AN/SRC-513
A238		C-51-574-000/MS-001 Technical Manual. Radio Set AN/SRC-514
A239		C-51-655-000/MB-001 MX2400 Integrated SATCOM Terminal Operator's Manual
A240		C-51-657-000/MS-001 Operation and Maintenance Instructions with Parts List - Satellite Communications Set AN/SC-3(V)17
A241		C-51-658-000/MS-001 TD-1271 B/U Multiplexer

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A242		C-51-659-000/MS-O01 Operation and Maintenance Instructions with Parts List, Organizational Level - Single DAMA Control Monitor
A243		C-51-661-000/MS-001 Operation and Maintenance Instructions UHF RT- 1107(V) 11/WSC-3(V)
A244		C-51-665-000/MS-001 Instruction Manual Power Amplifier RF-1110B (AN-5334/UR)
A245		C-51-666-000/MS-001 Operation and Maintenance Instructions, Radio Frequency switching SA-5147/UR
A246		C-51-667-000/MS-001 Operation and Maintenance Instructions with Parts Identification List, Indicator Operational Control ID5138/SR
A247		C-51-668-000/MS-001 Operation and Maintenance Instructions with Parts Identification List, Bandpass Filter Unit, F-5088/SR
A248		C-51-669-000/MS-001 Output Combiner Unit (CV-5165/UR Frequency Mixer Stage)
A249		C-51-670-000/MS-001 Instruction Manual Control Transmitter CB-5021/UR
A250		C-51-678-000/MS-001 Instruction Manual Instruction Manual Filter Bypass, F-5082A/URC
A251		C-51-693-000/MB-001 RF-5000 Falcon HF Communication System
A252		C-51-711-000/ME-001 Spectra VHF Radio
A253		C-51 -806-000/MS-001 Technical Manual Operation and Maintenance Instructions, Operational Radio frequency Conditioning Unit (commonly known as RFCU) HFDU-6130-2
A254		C-51-808-000/MB-001 Operators Manual for MX2464 Magnaphone B Integrated SATCOM Terminal
A255		C-51-81-000/MS-001 Technical Manual Miniloop Tunable Antenna Model MLA-115. 1.8-15 MHZ
A256		C-51-811-000/MS-001 Intermediate Maintenance Manual CD98/URC, Receive Mute Matrix Unit (RMMU)
A257		C-51-812-000/MS-001 Operators Manual, Control/Monitoring Subsystem
A258		C-52-000-000/AX-000 Naval Alarm, Signalling, Interior Comm Index
A259		C-52-101-000/MS-000 AN/SIH-503 Public Address System Instruction Manual
A260		C-52-106-XX/XXX AN/PIC-2 Public Address Set

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A261		C-52-239-000/MS-001 AN/SIA-120B (U) Amplifier Control Group
A262		C-52-257-000/MM-000 Integrated Central Broadcast System
A263		C-52-263-000/MS-000 Integrated Central Broadcast System and Audio Broadcast System
A264		C-52-394-000/MS-001 AN/SSC-512 (V) Trump SHINCOM
A265		C-52-394-000/MY-001 AN/SSC-512 (V) Trump SHINCOM
A266		C-52-415-000/MF-001 J-5222/SIC Comm Equipment Interface Unit and PP-S387 SIC
A267		C-52-448-000/MB-001 Operators Handbook, SHINCOM 2100
A268		C-52-448-000/MS-001 Description and Maintenance Instructions for SHINCOM 2128 System (IRO)
A269		C-53-279-000/MN-000 AN/VRC Radio Set
A270		C-53-279-F00/MS-000 AS-5041/URC Ultra High Frequency AE
A271		C-54-000-000/AX-000 Fixed Comm Equipment Index
A272		C-55-000-000/AX-000 Microwave and Visual Comm Index
A273		C-57-118-000/MB-000 AN/URN-20A (V1 TACAN
A274		C-57-118-000/MB-001 AN/URN-20A (V)1 Radio Set Modified
A275		C-57-118-000/ME-001 AN/URN-20A (V)1 Radio Set
A276		C-57-382-000/MS-001 Sonar Sounding Set UQN-4
A277		C-57-540-000/MB-001 Omega MX1105 Navigators Manual
A278		C-57-546-000/MS-001 OE-273 (V) URN Antenna Group for TACAN
A279		C-57-546-000/MS-002 OE-273 (V) URN Antenna Group for TACAN
A280		C-57-608-000/MS-001 Navigation Set, Loran-C/GPS AN/SRN-501
A281		C-57-610-000/MS-001 AN/SN-S Technical Manual
A282		C-57-611-001/MS-001 Doppler Speed Log System
A283		C-57-627-000/MS-001 Technical Manual INDB AN/SYN-503
A284		C-57-627-000/MS-002 Technical Manual INDB AN/SYN-SO3

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A285		C-57-686-000/MS-001 Inspection and Operation Instructions Satellite emergency Position Maintenance Beacon (EPIRB) Kannad 506
A286		C-57-689-000/MB-001 Navigational Computer AN/USN-502
A287		Superceded by A259
A288		C-57-689-000/MS-001 Installation and Maintenance Manual With Parts List, Inertial Navigation Set AN/SN-502
A289		C-57-689-000/MS-002 Inertial Navigation Set AN/SN-502
A290		C-57-732-000/MB-001 Inspection and Operation Instructions for Radar Transponder Rescuer SART
A291		C-57-735-000/MS-001 Naval GPS Navigation System (Trimble) Book 1 of 2
A292		C-57-735-000/MS-001 Naval GPS Navigation System (Trimble) Book 2 of 2
A293		C-57-787-000/MS-001 SHINNADS
A294		C-57-799-000/CS-001 Special Maintenance Instructions, SQS-510 Sonar Set
A295		C-58-144-000/MJ-001 AN/ULQ-6A Counter Measures Set
A296		C-58-144-000/MB-000 AN/ULQ-6A Counter Measures Set
A297		C-58-144-000/MS-001 AN/ULO-6C Counter Measures Set
A298		C-58-145-000/CS-001 Special Information Instructions
A299		C-58-146-000/MJ-000 AN/SLA-15 Antenna Group for ULQ
A300		C-58-146-000/MS-001 Technical Manual Volume 1
A301		C-58-146-000/MS-001 Technical Manual Volume 2
A302		C. 58-251-000/MD-001 AN/SLO-501 RAMSES Description (Confidential)
A303		C-58-252-000/MM-001 AN/SLA-10B Blanker-Video Mixer Group
A304		C-58-258-001 /MS-001 Shield II Type 210 Chaff and IR Decoy System
A305		C. 58-258-001/MS-002 Shield II Type 210 Chaff and IR Decoy System
A306		C-58-376-000/MS-001 AN/SRD-503 DF Set Operation and Maintenance
A307		C-59-010-009/MS-000 AN/UPA-59A (V) Operation and Maintenance Manual with parts list

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A308		C-59-010-009/MS-001 AN/UPA-59A & 59B Decoder Group
A309		C-59-107-000/MS-000 Control Transponder Set
A310		C-59-107-000/MS-001 C620/APX Operation and Maintenance Manual with parts list
A311		C-59-667-000/MS-001 Receiving Set Radio, AN/ARR-75
A312		C-59-667-000/MS-003 Sonobuoy Processing System (SPS)
A313		C-59-747-000/MD-001 AN/SPS-501 System Description LW-08
A314		C-59-756-000/MD-001 AN/SPS-502 System Description DA-08
A315		C-59-678-000/MS-001 AN/SPX-501 (V)
A316		C-59-678-000/MS-Z01 Supplement for CP/TRUMP Systems Operation and Maintenance Instructions
A317		C-59-678-000/MS-Z02 AN/SPX-501 VI3
A318		C-59-680-000/MF-000 AN/TPX-54 (V) RX/TX Group Interrogator
A319		C-59-680-000/MF-001 OR5014/TPX-54 (V) RX/TX Group Interrogator
A320		C-59-680-000/MS-001 TS-5144/SPX-501 (V) and TS-5145 Operation & Maintenance Instructions
A321		C-59-680-000/MF-Z10 Technical Manual Supplement IFF1
A322		C-59-680-000/MS-001 IF Test Set Loop
A323		C-59-681-000/MS-001 MK12 IFF RX/TX Technical Manual
A324		C-59-747-000/MD-001 LW-08 Part 1; System Description
A325		C-59-747-000/MD-Z01 LW-08 System Description Supplement
A326		C-59-747-000/MD-002 LW-08 Part 2; Technical Description
A327		C-59-747-000/MB-001 LW-08 Part 3; Operating Instructions
A328		C-59-747-000/AX-001 LW-08 Part 8; Catalogue of General Information
A329		C-59-755-000/D-001 ATMS Part 1; System Description
A330		C-59-755-000/MD-002 ATMS Part 2: Technical Description
A331		C-59-755-000/MB-001 ATMS Part 3; Operating Instruction

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A332		C-59-756-000/MD-001 DA-08 Part 1; System Description
A333		C-59-756-000/MD-Z01 DA-08 Part 8; System Description Supplement
A334		C-59-756-000/MD-002 DA-08 Part 2; Technical Description
A335		C-59-756-000/MB-001 DA-08 Part 3; Operating Instruction
A336		C-59-756-000/AX-001 DA-08 Part 8: Cataloque of General Information
A337		C-59-756-000/MB-001 DA-08 Part 3; Operating Instruction
A338		C-59-756-000/AX-001 DA-08 Part 8; Catalogue of General Information
A339		C-59-755-000/MD-001 ATMS Part 1: System Description
A340		C-59-755-000/MD-002 ATMS Part 2; Technical Description
A341		C-59-755-000/MB-001 ATMS Part 3; Operating Instruction
A342		C-59-755-000/MQ-001 ATMS Part 6B4; Computer Message Formats
A343		C-59-755-000/AX-001 ATMS Part 8; Catalog of General Information
A344		C-59-756-000/MD-001 LW-08 Part 1; System Description
A345		C-59-756-000/MD-Z01 LW-08 Part 1; System Description Supplement
A346		C-59-756-000/MD-002 LW-08 Part 2; Technical Description
A347		C-59-756-000/MB-001 LW-08 Part 3; Operating Description
A348		C-59-756-000/AX-001 LW-08 Part 8; Catalog of General Information
A349		C-59-770-000/MS-001 AS 177 Antenna
A350		C-59-795-000/MS-001 Sonobuoy Processor Sub-system SPSS3-1
A351		C-59-849-000/MS-001 Pathfinder Instruction Manual Vol. 1
A352		C-59-849-000/MS-002 Pathfinder Instruction Manual Vol. 2
A353		C-51-530-000/MS-001 Receiving Set, Satellite Signal AN/SSR-1 and AN/SSR-1A

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A354		C-51-530-000/MB-001 Receiving Set, Satellite Signal AN/SS-1 and AN/SSR-1A
A355		C-60-162-000/MS-001 Radar Indicator Group AN/SPA-25F
A356		C-60-162-000/MS-002 Radar Indicator Group AN/SPA-25F
A357		C-60-179-000/MS-001 AN/SYQ-503 RDDS Operating and Maintenance Instruction
A358		C-60-179-000/MS-001 AN/SYQ-503 RDDS Technical Manual Vol. I
A359		C-60-179-000/MS-001 RDDS Technical Manual Vol. 11
A360		C-60-211-000/MS-002 Technical Manual Operational and Maintenance Instructions MX-512-PV/RC Remote Control Link Monitor
A361		C-63-000-000/AX-001 Test Equipment Index General Purpose
A362		C-63-278-000/MS-001 Technical Manual for HF/M Receiving Antenna Multicoupler Model AMC/21
A363		C-64-376-000/MS-001 Fire Veto Panel
A364		C-67-R00-000/MS-002 AN/UPM-504 (V) Radar Test Set
A365		C-67-R00-000/MS-Z01 Test Set Supplement
A366		C-69-125 000/MS-001 Sonar AN/SQS-505 (V)
A367		C-69-462-000/MS-000 AN/SQA 502 Sonar Hoist Mechanism Handbook
A368		C-69-675-B00/MG-000 Test Set Assemblv MK432
A369		C-69-675-A00/MG-000 Control Panel MK 264
A370		C-69-676-000/MG-002 SVTT MK 32 MOD 5
A371		C-69-690-000/MS-001 Transmitter Group Heading Reference Data OT-5002/AN/SQS-505 (V)
A372		C-69-700-000/MS-000 Communication Set Sonar AN/WQC-501 (V) UWT
A373		C-69-704-000/MS-001 Tow Condition Monitor
A374		C-69-713-000/MS-001 Transmitter Set Torpedo Countermeasures AN/SLQ-25 (U)
A375		C-69-770-000/MS-001 Torpedo Handling and Storage System
A376		C-69-766-000/MS-001 CS Hull Outfit Operator and Maintenance Instruction
A377		C-69-771-000/MG-002 SVTT MK 32 MOD 9
A378		C-69-799-000/MS-001 Sonar Set AN/SQS-510 System Maintenance Manual

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A379		C-70-110-000/MS-000 Wind Indicating Equipment Type F
A380		C-70-312-000/MD-001 Equipment Description Group MK73 MOD 1
A381		C-70-321-000/MS-001 CIWS MK 15 Mods 11 through 14
A382		C-70-325-000/MD-001 LIROD Part 1; System Description
A383		C-70-325-000/MD-Z01 LIROD Part 1A; System Description Supplement
A384		C-70-325-000/D-002 LIROD Part 2A; Technical Description Computer
A385		C-70-325-000/MD-003 LIROD Part 2B1: Technical Description Optronics
A386		C-70-325-000/MD-004 LIROD Part 2B2; Technical Descriotion Display
A387		C-70-325-000/MD-005 LIROD Part 2C; Technical Description Director
A388		C-70-325-000/MD-006 LIROD Part 2D1; Technical Description Radar
A389		C-70-325-000/MD-007 LIROD Part 2D2; Technical Description Radar Timing
A390		C-70-325-000/MD-008 LIROD Part 2E; Technical Description System Support
A391		C-70-325-000/CS-001 LIROD Part 8; Catalogue of General Information
A392		C-70-326-000/D-001 76/62 SRGM Technical Manual
A393		C-70-326-000/D-002 76/62 SRGM Technical Manual
A394		C-70-326-000/D-003 76/62 SRGM Technical Manual
A395		C-70-326-000/ME-001 76/62 SRGM Technical Manual
A396		C-70-326-000/NK-019 76/62 Lower Hoist Test
A397		C-70-327-000/MD-001 STIR Part 1; System Description
A398		C-70-327-000/MD-002 STIR Part 2A; Technical Description Computer Sub-system
A399		C-70-327-000/MD-003 STIR Part 2B; Technical Description Display
A400		C-70-327-000/MD-004 STIR Part 2C; Technical Description Director
A401		C-70-327-000/MD-005 STIR Part 2D1; Technical Description Radar
A402		C-70-327-000/MD-007 STIR Part 2E; Technical Description Weapons

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A403		C-70-327-000/D-008 STIR Part 2F; Technical Description System Support
A404		C-70-327-000/MB-001 STIR Part 3; Operating Instruction
A405		C-70-327-000/CS-001 STIR Part 8: Cataloque of General Information
A406		C-70-328-000/DC-001 Vertical Launching System MK41 MOD C
A407		C-70-328-000/MB-001 LCCP Computer Program Operator's Manual
A408		C-70-328-000/MD-001 Vertical Launching System MK41 MOD C
A409		C-70-330-000/MM-003 WI Sub-system
A410		C-70-330-000/MS-001 WI Operation/Technical Description
A411		C-70-330-000/MS-002 WI System Description
A412		C-70-336-000/MS-001 Fire Control Interface Unit
A413		C-70-353-000/NG-002 WDS/CCS Integration System
A414		EE130-BG-OMI-010 AN/USG64 (V) 7, AN/USG-64 (V) 8 SATCOM Control System Overview
A415		EE160-GP-OMI-010/W1 AN/USC-43 (V)ANDVT
A416		EE160-WA-OMP-010/W1 OK-454 (V) WSC Single DAMA
A417		ES701038 AN/SSC-512 (V) Terminal User's Instruction Supplement
A418		SPAR-ASG-AMN-1013 AN/SSC-505 (V) Interior Communications System Description and Maintenance Instructions
A419		MX2400 Magnavox Integrated SATCOM Terminal
A420		C-51-547-000/MS-001 UHF Radio Receiver-Transmitter RT-5085
A421		C-51-547-000/MS-002 UHF Radio Receiver-Transmitter RT-5085
A422		C-51-559-000/MF-001 Receiver Transmitter Radio RT-5073(VWSC-3 and RT-5073(V)1WSC-3
A423		C-51-568-000/MS-001 RF-590 Radio Receiver (R-5095B/G)
A424		C-51-576-000/MS-001 Ancillary Radio Equipment Group
A425		C-51-633-000/MS-001 Radio Transmitter T-5086A/URT (RF-1310-13)
A426		C-51-698-000/MS-001 Handbook - UHF antenna AS-5041/URC for radio set UHF/AN/VRC49

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A427		C-51-699-000/MS-001 RF-5710 HF Modem (MD-5077(V)
A428		C-51-756-000/MS-001 RF-2601 Fast Tune Automatic Antenna Coupler
A429		C-51-762-000/MS-001 RF-591A-01 Pre/Postselector
A430		C-51-810-A00/MS-001 Miniloop HF Tunable Antenna
A431		C-51-814-000/MS-001 Narrowband Transceiver Subsystem
A432		C-51-866-000/MS-001 Indicator Fault Locating ID-5139/UR
A433		C-51-870-000/MS-001 Transmit/Receive Switch RE-5062A/UR
A434		C-51-872-000/MS-001 Bandpass Filter F-5094/UR
A435		C-51-873-000/MS-001 Amplifier. Radio Frequency AM-5314A/URT
A436		C-53-279-000/MS-002 Radio Sets AN/VRC-49
A437		C-53-626-000/MB-000 RF-5200 Flacon Series Manpack Tactical Communication Systems Operations Manual
A438		C-54-962-000/MB-001 Modem Communications 2440
A439		C-56-560-000/MB-001 Modem Communications V.3224/V.3225
A440		C-57-118-000/MS-000 AN/URN-20A(V) TACAN Radio set
A441		C-57-118-000/MS-002 AN/URN-20A(V) TACAN Radio set
A442		C-57-118-000/MS-007 AN/URN-20A(V) TACAN Radio set
A443		C-58-244-000/MS-000 Drier, Air-Gas, Desiccant HD-5029/SRC
A444		C-59-803-ZAO/MS-001 CP-2205(P) (V)/USQ-125 Model No. MX-512PV Link-11 Data Terminal P/N 34000-()-() MX-512PV VME Card Set Link-11 Data Terminal P/N 33510-(-() MX-512PV VME DTS Module P/N 34168
A445		C-59-803-ZBO/MS-001 Link-11 Monitor System Single-Tone (LMS-11ST)
A446		957031-A SHINPADS Standard Display AN/UYQ-501(V)
A447		C-46-498-000/MS-001 Technical Manual for AN/UL-507 (V) Data Processing Set
A448		AN/USQ-69(V) Learner's Guide

REF #	NDID/DOCUMENT CONTROL NO	TITLE OF REFERENCE
A449		HALIFAX Class Combat System Users Manual
B		**(Non-CAF) Government of Canada Publications & Foreign Military (Non-DND) Publications that relate to this Qualification**
B01	ANEP 77	Naval Ship Code (NATO standard)
C		**Civilian Publications and Manuals that relate to this Qualification**
C01		PMX Doc. No. 101733100 Communication Control and Monitoring System User's Manual.
C02		PMX Doc. No. 101731809 User's Manual Maintenance Test Combat System Operational Test
C03		PMX Doc. No. 101731810 User's Manual Maintenance Test Combat System Alignment Test
C04		PMX Doc. No. 101731811 User's Manual Maintenance Test Combat Svstem Interface Test.
C05		User's Manual Maintenance Test Link 11 POFA Test.
C06		Underwater Acoustic System Analysis; Burdic, William S.; Prentice-Hall 1984
C07		Principles of Underwater Sound, 3rd ed.; Urick, Robert J.; McGraw-Hill 1983
C08		Principles of Naval Weapon Systems; Payne, Craig M.; Naval Institute Press 2006
C09		Radar Fundamentals; Faulconbridge, R. Ian; Argos Press (Scitech Publishing) 2002
C10		Communication Electronics, 2nd ed.; Frenzel, Louis E.; Glencoe/MacMillan/McGraw-Hill 1995
C11		Elements of Electromagnetics, 3rd ed.; Sadiku, Matthew N.O.; Oxford Press 2001
C12		Electronic Principles, 3rd ed.; Malvino, Albert P.; McGraw-Hill 1984
C13		Network+ Study Guide, 3rd ed.; Groth, David; Sybex 2001
C14		Industrial Hydraulics Manual, 2nd ed.: Vickers 1989
C15		Modern Control Systems, 8th ed.; Dorf, Richard C.; Bishop, Robert H.; Addison Wesley 1998
C16		Digital Systems, 5th ed.; Tocci, Ronald J.; Prentice-Hall 1991
D		**Other Publications and Manuals that relate to this Qualification**
D01		
D02		

NOTE: If additional references are identified during the TPWB this annex can be updated secretarially through NPTG HQ.

ANNEX B – TASK LIST

1. The task list includes all of the tasks associated with this qualification, and the outcome of the Train / No-train decisions. If a task is designated as requiring training, the PO where it is covered is indicated. If a task is designated as not requiring training, the justification for that designation is provided.

2. The following table defines some of the criteria used to support a no-train designation in the task list. Other reasons may be used if required.

CRITERIA	DEFINITION
A	**Previously qualified.** Tasks and knowledge already possessed by the student by CAF and/or RCN courses, training and/or experience.
B	**Will receive on future courses.** Tasks and knowledge will be covered as part of other CAF and/or RCN courses and/or training.
C	**Difficulty.** Task is not difficult to learn; formal training is not required.
D	**Importance.** Task is not important; it has no significant impact on mission success.
E	**Frequency.** Task is rarely performed, no formal training is required.
F	**Lack of equipment.** Currently there is no equipment in the CAF inventory to train this task.

Task #	Task Statement	Train / No Train	PO #	Substantiation/Comments

ANNEX C – QS SCALAR

A QS Scalar was not provided.

ANNEX D – RECORD OF DECISIONS QSWB

1. This record summarizes the conduct of the QSWB including major activities, important discussions and contentious issues, as well as all decisions and recommendations made by the Board. A description of the reasoning which supports a decision or recommendation is provided when required.

Title	**Naval Engineering Common QSPWB**			
Location of Board	**CFNES S-37 Rm B-15/B-17**			
Dates	**13 Feb – 09 Mar 2012**			
Board Composition				
	Surname	Initials	Rank	Unit
Chairperson	Tremblett	G	LCdr	CCFL HQ
DMAR Pers Rep	Day	B	LCdr	DMAR Pers
TDO	Prince	MMM	Capt	CFNES
SME	Stars	B	Lt(N)	DMSS 3
SME	Brideaux	DJ	Lt(N)	FMF CS Mech Eng
SME	Blais	RJV	Lt(N)	CFNES – OT Div
SME	Shillington	WA	Lt(N)	FMF CB Op
SME	Cyr	K	LCdr	MARPAC HQ N42
SME	Rector	I	Lt(N)	CFNES – OT Div
SME	Walker	G	Lt(n)	FMFCS / FTA
SME	Castelli	A	Lt(N)	PMO JSS
Administrative Assistant	Goodmurphy	J	NCdt	CFNES OT Div

BOARD OBJECTIVE:

1. A QSPWB was conduct in Oct 2010 and it became apparent that a Military Employment Structure Implementation Plan (MESIP) was required for the NTO MOSID in order to define clearly the job requirements. As waiting for the MESIP to be completed (3 to 5 years), a QSPRB was convened to bridge the gap of the current training.
2. Based on the QSPRB convening order, the QSPRB mandate was to:
   a. Separate the MS Eng BOQ and the NCS Eng BOQ QSPs common phase into one QSP called Naval Engineering Common. This will facilitate documentation management and will not affect training;
   b. Review and amend the existing POs, EOs and Assessment;
   c. Update the impact of the new equipment on the job requirements and the training; and
   d. Include “Risk Management” as a part of the NEC QSP to reflect the job requirements.

DISCUSSIONS / DECISIONS:

3. The Naval Technical Officer Training Management (NTO TM) at OT Div provided a general overview of the history of the NCS Eng/MS Eng courses

and the requirement to convene a QSPRB.

4. Based on the previous QSPWB (Oct 10), board objective and time allow to the QSPRB, the board members agree that the board will not focus on tasks and skill analysis.

5. The board members reviewed both QSPs and identified the common POs required in order to develop the NEC QSP. The following POs were removed:

a. From the MS Eng QSP;
   i. PO 012 – Manage Operational Readiness;
   ii. PO 013 – Manage Naval Maintenance;
   iii. PO 014 – Conduct Engineering Analysis;
   iv. PO 015 – Manage Naval Tests and Trials;
   v. PO 016 – Manage Training;
   vi. PO 017 – Lead Personnel;
   vii. PO 018 – Manage Projects;
   viii. PO 019 – Manage the Life Cycle of Materiel; and
   ix. PO 020 – Conduct Business Management.

b. From the NCS Eng QSP;
   i. PO 413 – Manage Operational Readiness;
   ii. PO 414 – Manage Naval Maintenance;
   iii. PO 415 – Conduct Engineering Analysis;
   iv. PO 416 – Manage Naval Tests and Trials;
   v. PO 417 – Manage Training;
   vi. PO 418 – Lead Personnel;
   vii. PO 419 – Manage Projects;
   viii. PO 420 – Manage the Life Cycle of Materiel; and
   ix. PO 421 – Conduct Business Management.

6. The following POs were developed in the new NEC QSP through the combination of the POs above in para 5:

a. PO 001 – Describe Operational Readiness;
b. PO 002 – Describe Risk Management;
c. PO 003 – Describe Naval Tests and Trials;
d. PO 004 – Support Engineering Analysis;
e. PO 005 – Describe Training and Personnel Management;
f. PO 006 – Describe Naval Materiel Management;
g. PO 007 – Describe Business Management; and
h. PO 008 – Manage Projects.

7. The following describes how each PO was created:

a. PO 001 – Describe Operational Readiness was created through the amalgamation of PO 012 – Manage Operational Readiness (MS Eng QSP) and PO 413 – Manage Operational Readiness (NCS Eng QSP). The majority of the material in the two POs was the same so the amalgamation was seamless;
b. PO 002 – Describe Risk Management is a new PO that was created in response to concerns raised by both Coasts. The

material was supplied by the CSEO Sea Training Pacific and was reviewed by the board members as the PO was developed;

c. PO 003 – Describe Naval Tests and Trials was created through the amalgamation of PO 015 – Manage Naval Tests and Trials (MS Eng QSP) and PO 416 – Manage Naval Tests and Trials (NCS Eng QSP). The majority of the material in the two POs was the same so the amalgamation was seamless;

d. PO 004 – Support Engineering Analysis was created through the amalgamation of PO 014 – Conduct Engineering Analysis (MS Eng QSP) and PO 415 – Conduct Engineering Analysis (NCS Eng QSP). The majority of the material in the two POs was the same so the amalgamation was seamless;

e. PO 005 – Describe Training and Personnel Management was created through the amalgamation of the two MSE POs, PO 016 – Manage Training and PO 017 – Lead Personnel, with the two CSE POs, PO 417 – Manage Training and PO 418 – Lead Personnel. There was a re-allocation of time with the PO to allow more time for the trainees to conduct divisional case studies. The remaining overall content between the MSE and CSE POs was similar so amalgamation was seamless;

f. PO 006 – Describe Naval Materiel Management was created through the amalgamation of two MSE POs: PO 013 – Manage Naval Maintenance and PO 019 – Manage the Life Cycle of Materiel, and two CSE POs: PO 414 – Manage Naval Maintenance and PO 420 – Manage the Life Cycle of Materiel. It was decided to delete DGMPR and DGPOR as the applicability with naval maintenance organizations is limited. It was also noted that the DMPOR functionality transferred to MARLANT N3. There were three changes to the overall PO content;

    i. Four periods were allocated for a tour of FMF Cape Scott;
    ii. Seven periods were added to the NDHQ tour to permit visits to industry; and
    iii. One period was added to provide a basic DRMIS introduction to include Purpose, Notifications and the Relationship between Orders and Notifications. This is in addition to the 7 day DRMIS course and should be provided prior to the 7 day course so that the trainee will have a basic understanding of DRMIS.

g. PO 007 – Describe Business Management was created through the amalgamation of PO 421 – Conduct Business Management (MS Eng QSP) and PO 421 – Conduct Business Management (NCS Eng QSP). The majority of the material in the two POs was the same so the amalgamation was seamless; and

h. PO 008 – Manage Projects was created through the amalgamation of PO 018 – Manage Projects (MS Eng QSP) and PO 419 – Manage Projects (NCS Eng QSP). This PO had originally 20 periods

allocated for Group Work on the projects, but the past course schedules had allocated 32. 12 periods were removed and used to offset the changes in time for PO 006. The remaining material in the two POs was the same so the amalgamation was seamless.

8. During the amalgamation of the MSE and CSE POs if time was different between the amalgamated POs, then the time for the new PO was defaulted to the higher time value.

9. The initial draft of the NEC QSP followed along the lines of the NCS ENG BOQ QSP. After some lengthy discussions, it was determined that the NEC QSP would follow the model of the MS ENG BOQ QSP. This entailed eliminating the EOs that defined the OJPR and placing them into the specific EO of that PO as separate teaching points.

10. During the early discussions, it was decided that there would be an OJPR included as part of this new QSP. It was also decided that this OJPR would be similar in format to the format that would be used in both the MS ENG BOQ OJPR and NCS ENG OJPR. It was also decided that, in order to ensure that it was more implicit for practice boards by the NTO Head of Department, there would be a requirement for practice board signature blocks for each PO that has an OJPR.

11. There was discussion over the length of the NEC material when it was included in the separate QSPs. It was stated that this was 20 days; however, the time allocated from each QSP was broken down as follows;

a. MS Eng for POs 012 to 020 is 25.71 days, assuming that 32 periods were allocated for project group work. This works out to 24 days if all time values are counted for exactly from the MS Eng QSP;
b. NCS Eng for POs 413 to 421 is 25.29 days, assuming that 32 periods were allocated for project group work. This works out to 23.57 days if all time values are counted for exactly from the NCS Eng QSP; and
c. NEC for POs 001 to 008 is 25.57 days for training, with 1 day admin and .5 days for the Critical EC, and then the NEC course length is 27.1 days.

12. It was decided after some discussion that the critical EC for NEC will remain as an online test with a minimum pass mark of 60%. Trainees will be given 3 periods to complete the test. OT Div will be responsible to manage the online test.

ASSESSMENT PLAN:

13. The assessment plan was updated to reflect the requirement for a critical EC on all of the NEC material and that it would also be testable during question 5 of the PC 001, the Qualification Board.

14. The Marking scheme is based on the discussion and decision made for the NCS Eng QSP as continuity. As part of the overall assessment of a trainee, the wording of the Chapter 3 paragraph of marking allocation was changed to the following:

a. In order to achieve a passing mark for the NEC ashore phase, trainees must achieve a minimum overall mark of 60%. The NEC ashore phase is represented by a critical EC mark and a formative mark. Trainees must achieve a minimum mark of 60% in the critical EC.
b. The final mark for the NEC is calculated using the following formula: Total Course Mark = 0.7x(Critical EC Mark) + 0.3x(Formative Mark)

LESSONS LEARNED:

15. The material in the new Risk Management PO will need to be re-visited after a couple of courses are taught.
16. A recommendation was made to remove PO 008 – Manage Projects from the NEC QSP and develop it as a separate QSP with its own qualification
17. It would be beneficial if a TDO and a chairman were assigned for each QSPs reviewed to guide adequately the board during the entire review board.

FOLLOW-UP ACTION:

SERIAL	ACTION ITEM	TARGET DATE	OPI
001	The CFNES OT Div Comd will need to discuss with the CSE Div Comd (MARPAC) that Board results will be by message and not letters.		OT Div Comd
002	Developed the Online critical EC		OT Div NTO
003	The reference to the MARCOM Production Books and the MARCOM Personnel Production and Scheduling Procedures in the NEC QSP Chapter One to be confirmed by the TDO		OT DIV TDO
004	Remove PO 008 – Manage Projects as a part of the NEC and develop as a separate QSP and get this course nationally accredited.		OT Div NTO Training Officer
005	OT DIV TDO to research new MITE code for NEC QSP		OT DIV TDO

RECORD OF AMENDMENT & AMPLIFICATION

1. The record of amendment and amplification will be filled in whenever a change to the QS is initiated. This section is used to provide amplification details and provide a history of the changes made. In the requirement for the amendment section, the member inputting the changes will transpose the information from the submitted QS Change Request (QSCR), thus ensuring the QSP has the most relevant information.

Version:	**2.0**
Date: 1 July 2023	**Who Initiated Change: HMCS Venture**
Requirement for the Amendment: New RCN QSP Format	
Description of Changes: QSP was edited to reflect the new RCN QSP standard. PO 006 had an EO added, EO 006.08 Describe Naval Material Assurance. This was added to reflect new Llyod's Registrar standards.	
Change Tracking number:	
Trg Mgr: Lt(N) Newcombe	

TRAINING PLAN ANNEXES

ANNEX E – RESOURCES, EQUIPMENT, SUPPLIES, AND ANCILLARIES

1. The following resources, equipment, supplies, and other elements are required in support of training. Resources are described in terms of formed organizations and are to arrive equipped with all personnel, vehicles, weapons, communications, and ancillary equipment required for conduct and integral support of operations described in the POs.

2. This information was not found in the previous QSP. This has to be entered in at a later date.

3. Facilities/Trainers

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				
NOTES:				

4. Ammunition

SER	LOG GUIDE	DESCRIPTION		QTY	REMARKS
a.					
b.					
c.					
NOTES:					

5. Weapons

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

6. Specialized/Technical Equipment

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

7. Transport/Vehicles

SER		NSN	DESCRIPTION		QTY	REMARKS
a.		XXX-XX-XXX-XXXX				
b.		XXX-XX-XXX-XXXX				
c.		XXX-XX-XXX-XXXX				
	NOTES:					

8. Communication Equipment/Frequencies

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

9. Training Aids

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				
NOTES:				

10. POL Products

SER	NSN	DESCRIPTION		QTY	REMARKS
a.	XXX-XX-XXX-XXXX				
b.	XXX-XX-XXX-XXXX				
c.	XXX-XX-XXX-XXXX				
NOTES:					

11. General Purpose Stores

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				
NOTES:				

12. Stationery

SER	DESCRIPTION		QTY	REMARKS
a.				
b.				
c.				
NOTES:				

13. Personal Equipment

All students will have in their possession the personal equipment detailed in the course joining instructions. All personal equipment to be assigned to the students are accounted for in the resource tables above.

ANNEX F – COSTING INFORMATION

This information was not found in the previous QSP. This has to be entered at a later date.

Estimates only, based on CF Cost Factors Manual

cost-factors-manual (mil.ca)

****5 year total based on one course per year INCLUDING Initial Set-Up. Costs may be adjusted accordingly.**

Course: EXAMPLE		Serial:				
TD	No. Of Students / instructors /	Cost per each	# Days (if applicable)	Initial TE Set-up	Each successive course	5 Year total
Subtotal						
Material	Quantity	Cost per each	# Days (enter 1 if not applicable)			
Subtotal						
PYs (only if course includes incremental on SWE, or course to be delivered off site	Quantity	Cost per day* (SWE if necessary)	# Days (if applicable)			
	Total Estimated Costs					

ANNEX G – SUGGESTED SEQUENCING AND TIMINGS

1. This annex is to be used to properly schedule all required classrooms, trainers, ships, personnel, etc. for the duration of the course. Admin and PT periods are accounted for IAW NPTORD 4500-22 Annex B (QSP Instructions).

2. The tables are completed in the preferred sequence for lessons and resources. Special instructions to schedulers shall be included as "Notes" following the applicable PO or Module.

3. The table below is available to use as a template for calculations.

Allocation of ATT	**Minutes**
Actual Training Time	8800
ATT Sub-total	
Allocation of BAT, TAT, TTT, PT (Course > 31 days)	
Base Administrative Time (BAT)	0
Training Administrative Time (TAT)	500
Training Travel Time (TTT)	800
Physical Fitness (PT)	750
Sub-total	
Course Duration Calculation (ATT+BAT+TTT+TAT+PT) **Total mins/400 = # days** (rounded up)	
Total Training Days	**27.5**

OPTIMUM TRAINING SCHEDULE

Week 1 **Time**	**Day 1**	**Day 2**	**Day 3**	**Day4**	**Day 5**
Period 1 **0800-0850**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 2 **0900-0950**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 3 **1000-1050**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 4 **1100-1150**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 5 **1300-1350**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 6 **1400-1450**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 7 **1500-1550**	**EO/EC /PC/Admin** Title(s) / Activity				
Period 8 **1600-1650**	**EO/EC /PC/Admin** Title(s) / Activity				

ANNEX H – ASSESSMENT PLAN

EC / PC	Scope	Purpose	How	When	Time	Resources	Limitations	Pass/Fail Criteria	Impact
EO 008.01 – 008.03	The Eos Covered	The NTO shall demonstrate a detailed knowledge of Engineering Design and Integration	Group Project	At the completion of EOs 008.01 to 008.03	During the duration of the Eos	Materials budget, Wireless Controller	Nil	Pass; 60% on the material assessed	Trainees will be required to rectify any deficiencies
PC 001	Entire QSP	To evaluate the candidate's technical ability to perform as a Naval Combat Systems / Marine Systems Engineer	Oral Board	On completion of OJPR	Time is included in NCS Eng and MS Eng BOQ QSPs	Boardroom, board personnel in accordance with NAVORD 4500-20	Nil	Pass mark, achieve a minimum standard of 60%	First fail, reboard Second fail, TRB

ANNEX I – ASSESSMENT CHECKLISTS

Assessment checklists can be found in the trainee's respective QSPs for NCS ENG or MS ENG. (AIRY / AIRX)

ANNEX J – ON JOB PERFORMANCE RECORD (OJPR)

(Certificate of Qualification)

Version	Revision Date	Remarks
OJPR Version 2.0	01/07/2023	Formatted OJPR in new QSP Format

Name and Initials	Rank	Service Number

Date Training Started	
Date of Completion	

THE OJPR MUST BE COMPLETED IN ACCORDANCE WITH THE INSTRUCTIONS BELOW. INCORRECT OR INCOMPLETE OJPR WILL BE RETURNED FOR RE-SUBMISSION.

INSTRUCTIONS

1. One copy of this OJPR, hereinafter referred to as the master copy, shall be made up for each individual beginning OJT. It shall be kept and maintained by the supervisor. The Training Establishment (TE) shall make a duplicate copy for retention by the trainee for the purpose of recording and reviewing individual progress. The TE for the OJPR must maintain Training Tracker fields as outlined in Monitor Mass.

2. As the trainee completes each Performance or Enabling Check to the required standard, IAW the associated checklist in ANNEX I, the Training PO or Officer will date and sign the applicable item on the master and duplicate copies of the OJPR sheet and inform the individual of the action taken.

3. It may not be possible to conduct OJT for some POs if, for example, the necessary equipment is not fitted on a ship, is not on issue to the unit, or is otherwise unavailable. If the training PO or Officer encounters any of these issues they are directed to contact Standards at the appropriate TE for guidance.

4. If an individual demonstrates the ability to perform additional duties not specified in this QSP, an appropriate entry shall be made under "Additional Duties" at the end of the OJPR.

PROGRESS REVIEW

5. Progress review shall be carried out quarterly by the Divisional Officer and Training Officer.

PO/EO	Description	Remarks	Rank/Name/ Signature	Date DD/MM/YYYY	OPI
PO 001 Describe Operational Readiness					
EO 001.01	Describe Readiness and Policy Programs	Explain			HOD
EO 001.02	Describe Ship's State of Readiness	Participation			HOD
		Prepare and Staff			HOD
PO 003 Describe Naval Test and Trials					
EO 003.01	Describe Naval Tests and Trials Policies and Programs	Assist			HOD
		Report, Review and Advise			HOD
		Explain			HOD
PO 005 Describe Training and Personnel Management					
EO 005.02	Describe Personnel Management	Perform			HOD
		Explain			HOD
PO 006 Manage Naval Maintenance Onboard HMC Ships					
EO 006.03	Describe Naval Maintenance Procedures	Participate			HOD
		Plan and Support			HOD
EO 006.04	Describe Naval Maintenance Documentation	Assist			HOD
EO 006.05	Describe Life Cycle Materiel Management (LCMM)	Prepare and Staff			HOD

Progress Review Record

Authority & Period	Reviewed By	Signature	Date
Divisional Officer (Q1)			
Training Officer (Q1)			
Divisional Officer (Q2)			
Training Officer (Q2)			
Divisional Officer (Q3)			
Training Officer (Q3)			
Divisional Officer (Q4)			
Training Officer (Q4)			

TRAINING COMPLETION CERTIFICATE

I certify that_____has completed training to the standards stated in QSP – NEC, except as noted, and is recommended to challenge their respective Basic Officer Qualification board..

Date training completed:	
HOD	Date
XO	Date
CO	Date

CERTIFICATION OF QUALIFICATION

I certify that _____ has completed the OJT to the standards stated in the Naval Engineering Common QSP..

TE Cmdt/CO	Date

ANNEX K – TT PLAN

EO	COA 1 TEL Delivery	COA 2 Interim Delivery	**Impact if TEL not implemented** - Realized benefits - E/W Coast Delta

ANNEX L – RECORD OF DECISIONS TPWB

1. This record summarizes the conduct of the TPWB including major activities, important discussions and contentious issues, as well as all decisions and recommendations made by the Board. A description of the reasoning which supports a decision or recommendation is provided when required.

2. The information usually found in this annex can be found in Annex D, the previous writing board combined both the QSWB and TPWB.

RECORD OF AMENDMENT & AMPLIFICATION

3. The record of amendment and amplification will be filled in whenever a change to the TP is initiated. This section is used to provide amplification details and provide a history of the changes made. In the requirement for the amendment section, the member inputting the changes will transpose the information from the submitted TP Change Request (TPCR), thus ensuring the QSP has the most relevant information.

Version:	**2.0**
Date: 1 July 2023	**Who Initiated Change: HMCS Venture**
Requirement for the Amendment: Rewrite to new QSP Format	
Description of Changes: QSP Was rewritten to new RCN QSP format. New EO was created under PO 006, EO 006.08 Describe Naval Material Assurance. This was written to incorporate material from Lloyd's Registrar into Naval Engineering Common. No training time was added to the course, a review of the time and reduction of some periods throughout the course kept the total training time the same.	
Change Tracking number:	
Trg Mgr: Lt(N) Newcombe	

ANNEX M – ACRONYMS AND ABBREVIATIONS

1. This annex is to be utilized only to identify uncommon abbreviations used within this QSP.